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--- Summary ---

■ 1RT Correct values for
guests
1CA Total number for
grams for guest 1

1CA Teaspoons sugar
guest 1

1CA Total no of grams
for guest

1CA Teaspoons sugar
guest 2

1O Invalid (6) L4
M

(EC/NOVEMBER 2018) MATHEMATICAL LITERACY P2 5
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2.2.3

5M One mark for each
correct bar

(5) L2
D

[22]

QUESTION 3 [20]

Question Solution Explanation Topic
and
Level

3.1 3.1.1 Distance from Durban to Nelspruit = 689 km■

Speed = ■

■

$110 = 689$

■■■■■ ■

$110 \times \text{time} = 689$

Time = 689

110 ■

= 6,26 hrs ■

= 6hrs 16 min ■■

1RT Distance

1SF Substituting correct
values

1S Change subject of
the formula

1CA Answer in hours

1C Converting 6,26 hrs
to minutes

1CA Hours and min (6) L3

Maps

and

M

3.1.2 Ascending order

58; 273; 292; 342; 475; 532; 656; 1050; 1120;

1463 ■■

Median = 475 + 532

2 ■

= 503,5 ■

1RT Correct values

1M Ascending order

1M Concept of median

1CA Median

(4) L2

Maps

and

D

051015202530354045 Amount of sugar in grams
Different Drinks Amount of sugar in grams for
different drinks
6 MATHEMATICAL LITERACY P2 (EC/NOVEMBER 2018)
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3.1.3 Pretoria:
 $1120 + 273 + 342 + 292 + 532 + 58 + 1050 + 656 + 1463 + 475$
10
 $= 6261$ ■
10 ■
 $= 626,1 \text{ km}$ ■

Port Elizabeth: ■
■■■■■ + ■■■■■■ + ■■■■■■ + ■■■■■■ + ■■■■ + ■■■■■■ + ■■■■ + ■■■■ + ■■■■ + ■■■■
■■■
9 440
 $10 = 944 \text{ km}$ ■
Difference $= 944 - 626,1$ ■
 $= 317,9 \text{ km}$ ■ 1M Adding all correct
values

1M Divide by 10
1CA Mean

1RT Adding all correct
values
1CA Mean

1M Subtraction
1CA Difference (7) L2 and
L3
Maps
and D

3.1.4 Distance $= 998 \text{ km}$ ■
Cost $= 70 \times 998$ ■
 $= 69860 \text{ cents}$
 $= 698,60$
 $= R699$ ■
OR
 $998 \times 0,7$ ■
 $= 698,60$ ■
 $= R699$ ■ 1RT Correct distance
1M Multiplying by 70

1R Nearest rand

1RT Correct distance
1M Multiplying by 70

1R Nearest rand (3) L2
F and
Maps

[20]

QUESTION 4 [27]

Question Solution Explanation Topic
and
Level

$$\begin{aligned} 4.1 \text{ Area of circle} &= \pi \times r^2 \\ &= 3,142 \times 0,45 \times 0,45 \\ &= 0,636255 \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \text{Area of semicircle} &= \frac{\pi \times r^2}{2} \\ &= \frac{3,142 \times 4,9 \times 4,9}{2} \\ &= 37,71971 \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \text{Difference} &= 37,71971 - 0,636255 \\ &= 37,08 \text{ m}^2 \end{aligned}$$

1A Radius
1SF Substitution
1CA Area of centre

1CA Area of semi -circle

1CA Difference

NPR (5) L3

M

4.2.1 169 seats

3A No of seats (3) L2

Maps

4.2.2 4

15 1A Numerator

1A Denominator (2) L2

P

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4.2.3

$$\text{Right side} = 171$$

$$\begin{aligned} \text{Total amount} &= 171 \times 150 \\ &= R25\,650 \end{aligned}$$

$$\begin{aligned} \text{Left side} &= 1\,69 \times 150 \\ &= R25\,350 \end{aligned}$$

Middle = 255

Cost of 1 seat in the middle = 108

$$\begin{aligned} 100 \times 150 \\ = R162 \end{aligned}$$

$$\begin{aligned}\text{Total amount} &= 255 \times 162 \\ &= \text{R}41\,310 \blacksquare\end{aligned}$$

$$\begin{aligned}\text{Total amount for all seats} \\ &= 25\,650 + 25\,350 + 41\,310 \blacksquare \\ &= \text{R}92\,310 \blacksquare\end{aligned}$$

Statement invalid ■ CA from 4.2.1

1A No of seats on right side

1 CA Amount for right seats

1CA Amount for left seats

1A Amount for a middle seat

1CA Total amount for middle seats

1M Adding all values

1CA Total amount

1O Invalid (8) L4
F

$$\begin{aligned}4.3.1 \text{ Grade 11 boys} &= 191 - (54 + 73) \quad \text{OR } 137 - 73 \\ &= 191 - 127 \blacksquare = 64 \\ &= 64 \blacksquare\end{aligned}$$

$$\begin{aligned}\text{Total number of learners} &= 158 + 137 + 99 \\ &= 394 \blacksquare\end{aligned}$$

$$\% = 64$$

$$\begin{aligned}394 \times 100 \blacksquare \\ = 16,2\% \blacksquare\end{aligned}$$

1M Subtracting

1CA No.

■■ 2A

(2) DH

L2

$$\begin{aligned}3.3 \text{ Modal value} &= 15\,000 \blacksquare \\ 15\end{aligned}$$

$$\begin{aligned}100 \times 15\,000 \blacksquare \\ = \text{R}2\,250 \blacksquare\end{aligned}$$

$$\text{Year 1} = 105$$

$$\begin{aligned}100 \times 2\,250 \blacksquare \\ = \text{R}2\,362,50 \blacksquare\end{aligned}$$

$$\text{Year 2} = 105$$

$$\begin{aligned}100 \times 2\,362,50 \\ = \text{R}2\,480,63 \blacksquare \text{ 1RT Correct modal value}\end{aligned}$$

1M Calculated 15% of
modal value
1CA Value

1M Increased by

5%
1CA First year

1CA Final amount
L3 (6) DH&F

3.4 0 OR 0% OR Impossible ■■
2A Answer (2) DH
L2

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QUESTION 4 [31]

4.1 4.1.1 $A = 21\,168\,700 + 2\,305\,800 + 677\,000 + 2\,214\,400$
 $= 26\,365\,900$ ■

$B = 22\,165\,000$
 $27\,635\,900 \div 100$
 $= 80,2\%$ ■ 1 MA Added correct
values
1CA Total

1 Correct values
1CA % (4) DH
L2

4.1.2 Indian / Asian ■■ 2A Answer
(2) DH
L2

4.1.3 Two million two hundred and fourteen thousand four
hundred ■■ 2A Answer
(2) DH
L2

4.2 4.2.1 $57 \times 38 = 19$ ■ 1M Correct number of
seats on the left
1M Subtraction
1M Correct number of
seats on the right
1CA Answer (4) M&P
L2

4.2.2 Percentage of seats for the handicapped 4
 38×100 ■■

10,526315789

10,53% ■ 1RM Correct values
 numerator and
 denominator
 1M Multiplied by 100
 1CA Answer (3) M&P
 L3

4.3 4.3.1 Total distance = 35 + 57 + 11 + 21 + 59 + 41 ■
 = 224 km ■

Speed = ■■■■■■■■

■■■■■

105 = 224

■■■■■ ■

Time = 224

105

= 2,133 hours ■

Minutes = 0,133 × 60 ■

= 7,98

2 hours 8 minutes ■

Time of arrival = 7 :00 + 2 hours 8 minutes

= 9:08 am ■

Claim is valid ■ 1RM Correct values

1CA Distance

1M Substituting in
 formula

1 S Change subject of
 formula

1CA Time in hours

1C Time in minutes

1CA Arrival time

1O Claim valid (8) M,
 M&P

L3&4

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4.3.2

Amount before increase = 2,82

1,068 ■

= R2,64 ■

Total distance travelled = 224 x 2

= 448 km ■

Amount = 448 x 2,64 ■

= R1 182 ,72 ■

= R1 183 ■ CA from 4.3.1

1M Divided by 1,068

1A Amount

I CA Distance

1M Multiplied by
amount calculated

1CA Amount

1R Nearest rand (6) F
L3

4.3.3 R61 ■
R349 ■ 1A R61
1A R349 (2) M&P
L2

TOTAL: 100

NATIONAL
SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2016

MATHEMATICAL LITERACY P2
MEMORANDUM

MARKS: 100

Symbol Explanation

M Method

MA Method with accuracy

CA Consistent accuracy

A Accuracy

C Conversion

S Simplification

RT/RG/RM Reading from a table/Reading from a graph/Read from map

F Choosing the correct formula

SF Substitution in a formula

J Justification

P Penalty, e.g.

■ 1M Adding

45 min and

15 min

1A

1O Opinion

1A Adding
 1h57 and
 40 min
 1A Time to
 arrive at home
 10
 Conclusion (6)
 1.2.3 ■ Cleaning of the cinema ■
 ■ To prepare for the next showing ■
 OR
 ■ Allow the cinema crew to take a break ■
 Accept any other relevant reasons 1A First
 reason
 1A Second
 reason
 (2)
 1.3 1.3.1 Regular pricing = $75 + 55 \blacksquare\blacksquare = R130$ ■ 1A Identifying
 the correct
 values
 1M Adding
 1A (3)
 1.3.2 ■ It is too expensive on a Sunday or
 weekends ■■
 OR
 ■ On other days its much cheaper ■■
 Accept any other explanation 2O
 Explanation
 (2)
 [20]

(EC/NOVEMBER 2015) MATHEMATICAL LITERACY P2 3

Copyright reserved Please turn over QUESTION 2
 2.1 2.1.1 (a) Total cost (in rand) = $R200 \blacksquare + \text{number of minutes more than } 150 \blacksquare \times R0,60$ ■ 1A Rental
 1A Minutes
 more than 150
 1A Multiply by
 60 cents
 OR
 Total cost (in rand) = $R 200 \blacksquare + (\text{number of minutes more than } 150) \blacksquare \times R0,60$ ■ 1A Rental
 1A Minutes
 more than 150
 1A Multiply by
 60 cents
 OR
 Total cost (in rand) = $R200 \blacksquare + (\text{number of minutes} - 150) \blacksquare \times R0,60$ ■ 1A Rental
 1A Minutes
 more than 150
 1A Multiply by
 60 cents (3)
 (b) Total cost (in rand) = $R200 + (\text{number of minutes} - 150) \times R0,60$
 $= 200 + (625 - 150) \times 0,60$ ■
 $= 200 + (475 \blacksquare \times 0,60)$
 $= 200 + 285 \blacksquare$
 $= R485,00$ ■ CA from
 2.1.1 (a)
 1SF
 1S
 1S
 1A (4)

2.1.2

1 Mark for line

0–100

minutes

1 Mark for the
correct break -
even point

2 Marks for
any other 3
points plotted
correctly

1M Labelling

Call Package

A (5)

2.1.3 For the same number of minutes used ■ the same
amount will be paid for both Call packages ■ 1A refer to
minutes

1A refer to

cost (2)

2.1.4 Break -even point (450 ■; 380 ■) CA from 2.1.2

1CA 450

minutes

1CA 380 rand

If order

incorrect 0 (2)

Call package A

Call package B

4 MATHEMATICAL LITERACY P2 (EC/NOVEMBER 2015)

Copyright reserved Please turn over 2.1.5 Call package A

Total cost = $100 + (451 - 100) \times 0,80$ ■

= $100 + (351 \times 0,80)$

= $100 + 280,8$

= R 380,80 ■

Call package B

Total cost = $200 + (451 - 150) \times 0,60$

= $200 + (301 \times 0,60)$

= $200 + 180,60$

= R 380,60 ■

Difference = $R 380,80 - R 380,60$

= R 0,20 ■

OR

Call package A - Call package B

= $100 + (451 - 100) \times 0,80$ ■ - $200 + (451 - 150) \times$
0,60

= $100 + (351 \times 0,80) - 200 + (301 \times 0,60)$

= $100 + 280,80 - 200 + 180,60$

= R 380,80 ■ - R 380,60 ■

= R 0,20 ■

1SF

1CA

1CA

1CA
Difference

1SF

1CA
1CA
1CA

Difference (4)

2.1.6 Would recommend that Mr. Gardner use Call Package A ■

With R 300 on Call Package A he will get 350 minutes ■■

With R 300 on Call Package B he will get 316 minutes to 317 minutes ■■ 1CA Choosing

correct
package

2 CA

2 A (5)

2.2 Number of households = $10 \times 10 \times 10 \times 10$ ■
= 10 000 ■ 1A Identifying

10

1M Multiplying

10 four times

1CA (3)

[28]

(EC/NOVEMBER 2015) MATHEMATICAL LITERACY P2 5

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3.1 3.1.1 $A = 17,20 \times 1,14$ ■

= R 19,61 ■

B = 31,95 ■

1,14

= R 28,03 ■ OR $A = 17,20 + (17,20 \times 0,14)$ ■

= 17,20 + 2,41

= R 19,61 ■ 1M

1A

1M

1A (4)

3.1.2 No■, she will not pay for the first 6 kl ■ 1A

1O (2)

3.1.3 Payment for 19,5 kl

= $(6 \times 0) + (4,5 \times 7,60) + (9 \times 11,61)$ ■

= R 0 + R 34,20 + R 104,49 ■

= R 138,69 ■ $\times 1,14$

= R 158,11 ■ 1M

1S

1A

1A Including

VAT (4)

3.2 3.2.1 Diameter = 750 mm,

therefore radius = 375 mm ■ = 0,375 m

Height = 2230 mm = 2,230 m ■

Volume = $3,142 \times 0,375 \text{ m}^2 \times 2,230 \text{ m}$ ■

= $3,142 \times 0,140625 \text{ m}^2 \times 2,230 \text{ m}$

$$= 0,985311562 \text{ m}^3 \times 1000$$

$$= 985,311562 /$$

Accept 985,312

Therefore;

1000 l $\neq$ 985,311562 l OR 985,312 l 1A Finding radius

1A Convert

mm to m

(both)

1SF

1S radius²

1CA in litres

1A (6)

3.2.2 The given volume was rounded to the nearest

1000

Consumers are under the impression that you buy a tank with a capacity of 1 000 l, while it only has a tank capacity of 985 l. 1O

2O (3)

3.2.3 Tank can overflow if it is filled above the recommended filled height 2R (2)

3.2.4 Any of the vertical round water tanks 2A (2)

3.3 Bar graph and

Pie chart 1A Bar graph

1A Pie chart (2)

[25]

6 MATHEMATICAL LITERACY P2 (EC/NOVEMBER 2015)

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QUESTION 4

$$4.1.1.1 \text{ Mean} = \frac{26 + 29 + 22 + 23 + 22 + 21 + 24}{7}$$

$$= \frac{167}{7}$$

$$= 23,85714286$$

$$= 24^\circ\text{C}$$

1M adding all

day temp

1M/7

1CA

1CA (4)

$$4.1.2 \text{ Range} = 29^\circ\text{C} - 15^\circ\text{C}$$

$$= 14^\circ\text{C}$$

1M concept of

range

1A (2)

$$4.1.3 \text{ Median} = 15; 16; 17; 17; 17; 20; 21$$

$$= 17^\circ\text{C}$$

1M Arrange

values

1A (2)

4.1.4 Night temperatures decreases from Wednesday to Friday and then increases on Saturday and then from Sunday remains constant until Tuesday

2O (2)

4.1.5 Monday with

4 °C 1RT Day

1A Difference (2)

4.1.6 Friday 2RT (2)

4.1.7

Any 4 pairs

correctly
plotted
1Mark for
legend
1 Mark correct
graph (6)
4.2 It is highly unlikely that it will rain on Friday because it is
only 30% which is small chance for rain to fall ■■ 20 (2)
4.3 4.3.1 Time = Distance
Speed

= 64,7 km ■
90 km/h
= 0,7188....hours x 60 ■
= 43,133...
= 43 minutes ■

1M

1S

1A (3)
4.3.2 Your speed will decrease ■therefore it will take you
longer ■to complete the trip 1A refer to
speed
1A refer to
time (2)
[27]
TOTAL: 100
010203040
Wo Do VrSaSoMa DiTemperature in °C
Days Day and night
temperatures for George
from 14 Jan - 20 Jan
2015
Day temp
Night temp

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GRADE 1 1

NOVEMBER 2015

MATHEMATICAL LITERACY P1
MEMORANDUM

MARKS: 100

Symbol Explanation
M Method
A Accuracy
CA Consistent accuracy
RT/RG/RM Reading from a table/Reading from a graph/Read from map
RP Reading from the plan
SF Substitution in a formula
S Simplifications
P Penalty (no units, incorrect rounding off etc.)

2 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2015)
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 QUESTION 1

Question Solution Explanation Mark

1.1 1.1.1 $R169 \times 24 + R55 \times 24$ ■
 $R4\ 056 + R1\ 320$ ■
 $= R5\ 376$ ■ 2M
 1S
 1A (4)

1.1.2 $R0,79 \times 8$ ■
 $R6,32$ ■ 1M
 1S (2)

1.1.3 $R1\ 999 + R1\ 999 \times 4,4\%$ ■ OR $R1\ 999 \times 1,044$
 $= R1\ 999 + R87,96 = R2\ 086,96$
 $= R2\ 086,96$ ■
 OR
 $R1,999 \times 0,144$
 $R2\ 086,86$ 1M

1S
 (2)

1.2 1.2.1 $R0,35 \times 25$ ■
 $= R8,75$ ■ 1M
 1A (2)

1.2.2 $18:15$ ■
 $= 6:5$ ■ 1M
 1A (2)

1.2.3 $25 \div 5 = 5$ ■
 $35 \div 7$
 $= 0,71$ or 71% (Answer only – full marks) 1M
 1A
 (2)

1.3 1.3.1 Month 1: $R1\ 525 \times 20\%$
 $= R305,00$ ■
 Month 2: $R1\ 830,00 \times 20\%$
 $= R366,00$
 Interest $= R366 + R305 = R671,00$ ■
 1M

1S (2)

1.3.2 $R400 \times 11 \times 8 + R2\ 000 \times 8 + R671$ ■
 $= R35\ 200 + 16\ 000 + R671$
 $= R51\ 871,00$ ■ 2M
 1CA
 Multiplication
 by different
 number not 11 (3)

1.3.3 $R400 \times 11 + R2000 + R671$ ■ $= 400\ 000$
 $R4\ 400 + R2\ 000 + R671$ ■
 $= R7\ 071,00$ ■
 OR

$+ R671 = R7\ 071,00$ 1M

1S
1S
(3)

(EC/NOVEMBER 2015) MATHEMATICAL LITERACY P1 3

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Question Solution Explanation Mark

1.4 1.4.1 R4 760 x 11,82 ■ OR 4,760 x 11,82

= R56 263,20 ■ = 56,26 1M

1A (2)

1.4.2 R1 571,05 x 114% ■ OF R1 571,05 x 14%

R1 790,997 R219,95 ■

R1 790,997 x 2 ■ R1 571,05 + R219,95 ■

= R3 581,99 ■ R1 571,05 + R219,95 ■

OR

R1 571,05 x 14%

= R 219,95 ■

= R1 571,05 + R219,95 ■

= R1 790,997 x 2

= R3 581,99 ■

OR

R1 571,05 x 2 ■

= R3 142,10 x 114% ■

= R3 581,99 ■ 1M

1S

1A

(3)

[28]

4 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2015)

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QUESTION 2

Question Solution Explanation Marks

2.1 2.1.1 227 ■■ RD (2)

2.1.2 $7 \div 60$ ■ = 0,1166666667 h

$5 \div (60 \times 60) = 0,00138888888$ h ■

$4 \text{ h} + 0,1166666667 \text{ h} + 0,00138888888 \text{ h}$

= 4,118055556 hours

= 4 hours ■ 2M

1S

(3)

2.1.3 Average speed =

■

= 16,5 ■ km/hour ■ 1S

2S

Penalise for unit (3)

2.2 2.2.1 Arrival time – 18:56 ■

Departure time – 19:00 ■ 2RT

(2)

2.2.2 Bloemfontein ■■ 2RT (2)

2.2.3 Area of a bigger circle – Area of a smaller circle ■

$3,142 \times (5 \text{ m})^2 - 3,142 \times (2,5 \text{ m})^2$ ■

= 78,55 m² – 19,6375 m² ■

= 58,91 m² ■ (Accept 58,91 m²) ■ 1F

2SF

1S

1A (5)

[17]

(EC/NOVEMBER 2015) MATHEMATICAL LITERACY P1 5

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QUESTION 3

Question Solution Explanation

3.1 3.1.1 113 341 x 26% ■

= 113 341 x 0,26

= 29 468,66

= 29 469 ■

(2)

3.1.2 (i) Cancer ■

(ii) Heart disease ■

(2)

3.1.3

1 mark per three correctly plotted points

= 2 marks

2 marks for correctly labelling x -axis and y -axis

= 2 marks Types of

death Accidents Cancer Heart

disease Suicide Homicide All other

causes

% death 26 13 12 11 6 32 232

Causes of death ■

1 mark for completing the table

= 1 mark

1mark for joining points with a straight line

= 1 mark

(6)

3.1.4 113 341 – 39 213 ■

= 74 128 ■ 1M

1A (2)

3.1.5 25–44 years ■■ 2RG (2)

3.1.6 They drive recklessly trying to impress those people of their age about their driving skills.

2 WISKUNDIGE GELETTERDHEID V1 (NOVEMBER 2014)

Kopiereg voorbehou Blaai om asseblief

VRAAG 1

VR OPLOSSING Finansies

1.1.1 R70,17 x 3 + 2 x R2 367,54 + R28,06 x 3 + R28,94 ■ 1M

R210,51 + R4 735,08 + R84,18 + R28,94 ■ 1S

R5 058,71 ■ 1A (3)

1.1.2

■ 1M

= R1,40 ■ 1 A (2)

1.1.3 Desember ■■ 2RT (2)

1.1.4 BTW -nommer 8754896368 ■■ 2RT (2)

1.1.5 Snymasjien RYOBI 43CC ■■ 2RT (2)

1.1.6 Q 000002158 ■■ 2RT (2)

Meting

1.1.7 1 000 m■ = 1 ■

2 00 m■ = x ■ 1M

=

0,2 ■ = x ■

1A (2)

1.1.8 12 x 13 ■
= 156 maande ■ (2)

Data

Hantering

1.2.1 Skool E ■■ 2A (2)

1.2.2 Skool A fooie – Skool B fooie

R14 000 – R12 000 ■ 1M

= R2 000,00 ■ 1 A (2)

[21]

(NOVEMBER 2014) WISKUNDIGE GELETTERDHEID V1 3

Kopiereg voorbehou Blaai om asseblief

VRAAG 2

VR OPLOSSING Finansies

2.1.1 ■

■

2SF

= R 168 947,37 ■

OF

R192 600 x

R23 652,63■

R19 6200 – R23 652,63 ■

= R 168 947,37 ■ 1 A

OF

R192 600: x ■ = 114% : 100 ■

x = R168 947,37 ■

(3)

2.1.2

■ = R13,2 8 ■ 1D

1A (2)

2.2.1 Wins = R7 200,00 – R3 339,14 ■

= R3 860,86 ■ 1SF

1 A (2)

2.2.2 R350,00 x 4 = R1 400,00 ■ OF 1% of R350

R1 400,00 x 1% ■ = R3,50 x 4

= R14,00 ■ maand ■ = R14,00 1M

1S

1A (3)

$$\begin{aligned} 2.2.3 \quad & \quad \quad - \\ & \times 100 \blacksquare \\ & = \\ & \times 100 \blacksquare \end{aligned}$$

$$= 5,56 \text{ of } 5,6\% \blacksquare 1\text{SF}$$

1S

1CA
(3)

2.3.1

$\blacksquare$

=

$\blacksquare$

$$= 0,00074 \blacksquare 1\text{D}$$

1 S

1A

(3)

2.3.2 Paalement verskuldig =

$$+ R21,00 \blacksquare$$

$$\begin{aligned} &= R5\,753,05 \div 12 + R21,00 \blacksquare \\ &= R500,42 \blacksquare 1\text{SF} \end{aligned}$$

1S

1CA (3)

2.3.3 R6 879 $\blacksquare$

379,83 $\blacksquare$

$$= R\,18,1107 \blacksquare$$

$$= R18,11 \blacksquare 2\text{D}$$

1A

(3)

[22]

4 WISKUNDIGE GELETTERDHEID V1 (NOVEMBER 2014)

Kopiereg voorbehou Blaai om asseblief

VRAAG 3: ME TING

$$\begin{aligned} 3.1.1 \text{ Oppervlakte} &= 162 \text{ m} \times 160 \text{ m} \blacksquare \\ &= 25\,920 \text{ m}^2 \blacksquare 1\text{SF} \end{aligned}$$

1A (2)

$$\begin{aligned} 3.1.2 \text{ Omtrek} &= 2(160 \text{ m} + 162 \text{ m}) \blacksquare \\ &= 644 \text{ m} \blacksquare 1\text{SF} \end{aligned}$$

1A (2)

$$\begin{aligned} 3.1.3 \text{ Aantal gatte} &= 324 \div 2 - 1 \blacksquare \\ &= 162 - 1 \blacksquare \\ &= 161 \text{ gatte} \blacksquare 1\text{SF} \end{aligned}$$

1S

1A (3)

$$\begin{aligned} 3.1.4 \text{ BO} &= 2(6,2 \text{ m} \times 3 \text{ m}) + 2(5,1 \times 3 \text{ m}) - (1,71 \text{ m}^2 + 4,875 \text{ m}^2) \blacksquare \\ &= 37,2 \text{ m}^2 + 30,6 \text{ m}^2 - 6,585 \text{ m}^2 \blacksquare \\ &= 61,215 \text{ m}^2 \blacksquare \end{aligned}$$

1SF

1S

1A (3)

$$3.1.5 \blacksquare = 6 \text{ m}^2$$

$x = 61,215$ ■
 $61,215 \div 6 = x$
 $x = 10,2025 / 5$ ■
 $= 2,0405$
 Daarom 3 (5 ■) blikke verf ■ 1M

1M

1A (3)

3.2 ■

■ Penaliseer vir geen eenhede

$= 34,26 \text{ kg/m}^2$ ■■
 Aanvaar 114 kg ($33,97 \text{ kg/m}^2$) 1CV
 1SF
 1A
 1U (4)
 [17]

VRAAG 4

4.1.1 C2 ■■ 2RM (2)

4.1.2 Indiese oseaa n ■■ 2RM (2)

4.1.3 Swaziland ■ en Mosambie k ■ 2RM (2)

4.1.4 $6,5 \times 2600\,000 = 16\,900\,000$ ■
 $16\,900\,000 \div 100\,000$ ■
 $= 169 \text{ km}$ ■ 1M
 1C
 1A (3)

4.1.5 .Big Bend ■

.Lavumisa ■

.Mkuze■

.Phinda 3RM

(3)

(NOVEMBER 2014) WISKUNDIGE GELETTERDHEID V1 5
 Kopiereg voorbehou Blaai om asseblief
 4.1.6 Kwazulu -Natal ■■ 2A (2)

4.1.7 Qumbu ■
 Mount Frere ■ 2A
 (2)
 [16]

VRAAG 5: DATA HANTERING

5.1.1 Limpopo ■■ 2RT (2)

5.1.2 Gauteng : Totale aantal babas
 $60 : 320$ ■■
 : .

2 MATHEMATICAL LITERACY P1 (NOVEMBER 2014)
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QUESTION 1

Q SOLUTION Finance

$$\begin{aligned} 1.1.1 \quad & R70,17 \times 3 + 2 \times R2 \quad 367,54 + R28,06 \times 3 + R28,94 \quad \blacksquare \quad 1M \\ & R210,51 + R4 \quad 735,08 + R84,18 + R \quad 28,94 \quad \blacksquare \quad 1S \\ & R5 \quad 058,71 \quad \blacksquare \quad 1A \quad (3) \end{aligned}$$

$$\begin{aligned} 1.1.2 \\ \blacksquare \quad 1M \end{aligned}$$

$$= R1,40 \quad \blacksquare \quad 1A \quad (2)$$

$$1.1.3 \quad \text{December} \quad \blacksquare \blacksquare \quad 2RT \quad (2)$$

$$1.1.4 \quad \text{VAT number } 8754896368 \quad \blacksquare \blacksquare \quad 2RT \quad (2)$$

$$1.1.5 \quad \text{Brush cutter RYOBI 43CC} \quad \blacksquare \blacksquare \quad 2RT \quad (2)$$

$$1.1.6 \quad Q \quad 000002158 \quad \blacksquare \blacksquare \quad 2RT \quad (2)$$

$$\begin{aligned} & \text{Measurement} \\ 1.1.7 \quad & 1 \quad 000 \quad m \blacksquare = 1 \quad \blacksquare \\ & 2 \quad 00 \quad m \blacksquare = x \quad \blacksquare \quad 1M \\ & = \end{aligned}$$

$$0,2 \quad \blacksquare = x \quad \blacksquare$$

$$1A \quad (2)$$

$$\begin{aligned} 1.1.8 \quad & 12 \times 13 \quad \blacksquare \quad 1M \\ & = 156 \text{ months} \quad \blacksquare \quad 1A \quad (2) \end{aligned}$$

Data Handling

$$1.2.1 \quad \text{School E} \quad \blacksquare \blacksquare \quad 2A \quad (2)$$

$$\begin{aligned} 1.2.2 \quad & \text{School A fees} - \text{School B fees} \\ & R14 \quad 000 - R12 \quad 000 \quad \blacksquare \quad 1M \\ & = R2 \quad 000,00 \quad \blacksquare \quad 1A \quad (2) \\ & [21] \end{aligned}$$

(NOVEMBER 2014) MATHEMATICAL LITERACY P1 3
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QUESTION 2

Q SOLUTION Finance

$$\begin{aligned} 2.1.1 \quad & \blacksquare \\ & \blacksquare \quad 2SF \\ & 1A \quad (3) \end{aligned}$$

$$\begin{aligned} & = R168 \quad 947,37 \quad \blacksquare \\ & \text{OR} \\ & R192 \quad 600 \times \end{aligned}$$

$$\begin{aligned} & R23 \quad 652,63 \quad \blacksquare \\ & R19 \quad 6200 - R23 \quad 652,63 \quad \blacksquare \\ & = R168 \quad 947,37 \quad \blacksquare \end{aligned}$$

OR

$$R192\ 600 : \blacksquare = 114\% : 100 \blacksquare$$

$$x = R168\ 947,37 \blacksquare$$

2.1.2

$$\blacksquare = R13,28 \blacksquare 1D$$

1A (2)

$$2.2.1 \text{ Profit} = R7\ 200,00 - R3\ 339,14 \blacksquare$$

$$= R3\ 860,86 \blacksquare 1SF$$

1 A (2)

$$2.2.2 \ R350,00 \times 4 = R1\ 400,00 \blacksquare \text{ OR } 1\% \text{ of } R350$$

$$R1\ 400,00 \times 1\% \blacksquare$$

$$R3,50 \times 4$$

$$= R14,00 \text{ a month } \blacksquare$$

$$= R14,00 \ 1M$$

1S

1A (3)

2.2.3

$$\times 100 \blacksquare$$

$$=$$

$$\times 100 \blacksquare$$

$$= 5,56 \text{ or } 5,6\% \blacksquare 1SF$$

1S

1CA

(3)

2.3.1

$$\blacksquare$$

$$=$$

$$\blacksquare$$

$$= 0,00074 \blacksquare 1D$$

1 S

1A

(3)

2.3.2 Instalment due =

$$+ R21,00 \blacksquare$$

$$= R5\ 753,05 \div 12 + R21,00 \blacksquare$$

$$= R479,42 + R\ 21,00$$

$$= R500,42 \blacksquare 1SF$$

1S

1CA

(3)

$$2.3.3 \ R6\ 879 \blacksquare$$

$$379,83 \blacksquare$$

$$= R18,1107 \blacksquare$$

$$= R18,11 \ 2D$$

1A

(3)

[22]

4 MATHEMATICAL LITERACY P1 (NOVEMBER 2014)

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QUESTION 3: MEASUREMENT

$$3.1.1 \text{ Area} = 162 \text{ m} \times 160 \text{ m} \blacksquare$$

$$= 25\ 920 \text{ m}^2 \blacksquare 1SF$$

1A (2)

$$3.1.2 \text{ Perimeter} = 2(160 \text{ m} + 162 \text{ m}) \blacksquare \\ = 644 \text{ m} \blacksquare 1\text{SF}$$

1A (2)

3.1.3 No.

--- Full Combined Text ---

NATIONAL
SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2018

MATHEMATICAL LITERACY P2
MARKING GUIDELINE

MARKS: 100

Symbol Explanation

M Method

MA Method with accuracy

CA Consistent accuracy

A Accuracy

C Conversion

S Simplification

RT/RG/RM Reading from a table/Reading from a graph/Reading from a map

F Choosing the correct formula

SF Substitution in a formula

J Justification

P Penalty, e.g. for no units, incorrect rounding off etc.

R Rounding Off/Reason

AO Answer only

NPR No penalty for rounding

This marking guideline consists of 7 pages.

2 MATHEMATICAL LITERACY P2 (EC/NOVEMBER 2018)

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QUESTION 1 [31]

Question Solution Explanation Topic
and

Level

$$1.1.1 \text{ Washing and drying} = 18 \times 16 \\ = \text{R}288 \blacksquare$$

$$\text{Staysoft} = 5 = 3 \\ 18 = 3$$

$$1 \times 18$$

$$5 \blacksquare$$

$$= \text{R}10,80 \blacksquare$$

$$\text{Ironing} = 5 = 20 \\ 18 = 20$$

$$1 \times 18$$

$$5$$

$$= \text{R}72 \blacksquare$$

$$\text{Total} = 288 + 10,80 + 72$$

$$= \text{R}370,80 \blacksquare 1 \text{ MA Amount for} \\ \text{washing and drying}$$

1M Calculating
Staysoft
1CA Cost for Staysoft

1CA Cost for Ironing

1CA Total (5) L3
F

1.1.2 (a) Helper amount

No. of days = 9 ■

Amount = 9×150
= R1 350 ■

Wash and dry = $15 \times 5 \times 16$
= R1 200 ■

Stay soft = $5 = 3$
 $75 = 3$

1×75
5
= R45 ■

Handwash = $2 \times 5 \times 25$
= 250 ■

Total amount for laundry = $250 + 45 + 1200$
= R1 495 ■

Difference = $R1\ 495 - R1\ 350$
= R145 ■

Invalid laundry more expensive ■ 1A No. of days

1CA Amount for helper

1CA Amount for wash
and dry

1CA Staysoft

1CA Hand wash

1CA Total amount

1CA Difference

1O Invalid (8) L4
F

1.1.2 (b) Probability = $3/31 \times 100$ ■■

= 9,7% ■ 1A Numerator

1A Denominator

1CA Answer as a % to
1 decimal place (3) L2
P

(EC/NOVEMBER 2018) MATHEMATICAL LITERACY P2 3
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1.2.1 Across the length = 3 ■
0,5 ■

$$= 6$$

Across the breadth = 2,5
0,5

$$= 5 \blacksquare$$

Number of containers fitting = 6×5
= 30 ■

10 can fit ■ 1M Dividing
1C Conversion to metres

1CA Breadth/ Width

1CA Number of containers
1O Opinion (5) L3
Maps

1.2.2 (a) Volume of container
 $V = \pi \times (\text{radius})^2 \times \text{height}$
= $3,142 \times 25 \text{ cm} \times 25 \text{ cm} \times 70 \text{ cm}$ ■
= 137 462,5 cm³
= 137 462,5 cm³ ■

No. of litres = $137\,462,5 \div 1\,000$
= 137,4625 litres ■

1 Calculating radius

1CA Volume

1C Litres
NPR (3) L3
M

1.2.2 (b) Capacity for 5 containers
= $137,4625 \text{ litres} \times 5$
= 687,3125 litres ■

Water for 5 days of hand wash
Washing = 30×5
= 150 kg ■

25 kg = 100 litres
150 = 100
 1×150
25 ■
= 600 litres ■

Statement is valid ■ CA from 1.2.2 (a)
1MA Litres for 5 containers

1MA Amount of washing

1M Number of litres

1CA Total litres

1 O Valid (5) L4

M

1.2.3 Some material cannot be washed using machine . ■■

OR

Clothes can be damaged ■■

Accept any other relevant reason . 2A Reason or explanation

(2) L4

D

[31]

4 MATHEMATICAL LITERACY P2 (EC/NOVEMBER 2018)

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QUESTION 2 [22]

Question Solution Explanation Topic
and

Level

2.1 Amount for editing = 97×100 ■
= R 9 700 ■

Typist = ■■■■ ■■■■

$60 \times \text{rate} \times \text{no. of documents}$
= 28

$60 \times 195 \times 100$ ■■
= R9 100 ■

% = 105

$100 \times 9\,100$ ■ OR 9 700

$9\,100 \times 100$

= R9 555 ■ = 106,59% – 100%
= 6,59%

Not valid, more than 5% ■ 1M Multiply correct
values

1A Amount for editing

1SF Substituting

1A Correct values

1CA Amount for typist

1M Increasing by 5%

1CA Increased amount

1O Invalid (8) L4

F

2.2 2.2.1 31,7; 33,7; 35; 36,3; 36,4; 37; 39,2; 40,6; 41; 42,3;
44,1

Range = $44,1 \text{ g} - 31,7 \text{ g}$ ■■

= 12,4 g ■ 1RT Reading the correct

high and low values .

1M Subtraction

1CA Range (3) L2

D

2.2.2 Guest 1: Play energy drink = 29,2 g

Powerade = 38,8 g ■

Total = 68 g ■

Guest 1: 5g = 1 teaspoon

■ 68 = 1

1×68

5

= 13,6 teaspoons ■

Guest 2: Appletiser = 31,7g

Coca-Cola = 35g

Total = 66,7 g ■

Guest 2: 66,7

5

= 13,34 teaspoons ■

Statement not valid. Energy drink contains more sugar. ■ 1RT Correct values for

guests

1CA Total number for
grams for guest 1

1CA Teaspoons sugar
guest 1

1CA Total no of grams
for guest

1CA Teaspoons sugar
guest 2

1O Invalid (6) L4
M

2.2.3

5M One mark for each correct bar

(5) L2
D

[22]

QUESTION 3 [20]

Question Solution Explanation Topic and Level

3.1 3.1.1 Distance from Durban to Nelspruit = 689 km■

Speed = ■

■

110 = 689

■■■■■ ■

110 × time = 689

Time = 689

110 ■

= 6,26 hrs ■

= 6hrs 16 min ■■

1RT Distance

1SF Substituting correct values

1S Change subject of the formula

1CA Answer in hours

1C Converting 6,26 hrs to minutes

1CA Hours and min (6) L3

Maps

and

M

3.1.2 Ascending order

58; 273; 292; 342; 475; 532; 656; 1050; 1120;

1463 ■■

Median = 475 + 532

2 ■

= 503,5 ■

1RT Correct values

1M Ascending order

1M Concept of median
1CA Median
(4) L2
Maps
and
D

051015202530354045 Amount of sugar in grams
Different Drinks Amount of sugar in grams for
different drinks
6 MATHEMATICAL LITERACY P2 (EC/NOVEMBER 2018)
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3.1.3 Pretoria:
 $1120 + 273 + 342 + 292 + 532 + 58 + 1050 + 656 + 1463 + 475$
10
 $= 6261$ ■
10 ■
 $= 626,1 \text{ km}$ ■

Port Elizabeth: ■
■■■■■ + ■■■■■ + ■■■■■ + ■■■■■ + ■■■■ + ■■■■■ + ■■■■ + ■■■■ + ■■■■ + ■■■■
■■■
9 440
 $10 = 944 \text{ km}$ ■
Difference $= 944 - 626,1$ ■
 $= 317,9 \text{ km}$ ■ 1M Adding all correct
values

1M Divide by 10
1CA Mean

1RT Adding all correct
values
1CA Mean

1M Subtraction
1CA Difference (7) L2 and
L3
Maps
and D

3.1.4 Distance $= 998 \text{ km}$ ■
Cost $= 70 \times 998$ ■
 $= 69860 \text{ cents}$
 $= 698,60$
 $= R699$ ■

OR
 $998 \times 0,7$ ■
 $= 698,60$ ■
 $= R699$ ■ 1RT Correct distance
1M Multiplying by 70

1R Nearest rand

1RT Correct distance

1M Multiplying by 70

1R Nearest rand (3) L2

F and

Maps

[20]

QUESTION 4 [27]

Question Solution Explanation Topic

and

Level

$$\begin{aligned} 4.1 \text{ Area of circle} &= \pi \times r^2 \times \pi \\ &= 3,142 \times 0,45 \times 0,45 \\ &= 0,636255 \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \text{Area of semicircle} &= \pi \times r^2 \times \pi \\ &= 3,142 \times 4,9 \times 4,9 \\ &= 37,71971 \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \text{Difference} &= 37,71971 - 0,636255 \\ &= 37,08 \text{ m}^2 \end{aligned}$$

1A Radius

1SF Substitution

1CA Area of centre

1CA Area of semi -circle

1CA Difference

NPR (5) L3

M

4.2.1 169 seats

3A No of seats (3) L2

Maps

4.2.2 4

15 1A Numerator

1A Denominator (2) L2

P

(EC/NOVEMBER 2018) MATHEMATICAL LITERACY P2 7

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4.2.3

Right side = 171

$$\begin{aligned} \text{Total amount} &= 171 \times 150 \\ &= R25\,650 \end{aligned}$$

$$\begin{aligned}\text{Left side} &= 169 \times 150 \\ &= 25\,350 \blacksquare\end{aligned}$$

$$\text{Middle} = 255$$

$$\text{Cost of 1 seat in the middle} = 108$$

$$\begin{aligned}100 \times 150 \\ &= R162 \blacksquare\end{aligned}$$

$$\begin{aligned}\text{Total amount} &= 255 \times 162 \\ &= R41\,310 \blacksquare\end{aligned}$$

$$\begin{aligned}\text{Total amount for all seats} \\ &= 25\,650 + 25\,350 + 41\,310 \blacksquare \\ &= R92\,310 \blacksquare\end{aligned}$$

Statement invalid ■ CA from 4.2.1

1A No of seats on right side

1CA Amount for right seats

1CA Amount for left seats

1A Amount for a middle seat

1CA Total amount for middle seats

1M Adding all values

1CA Total amount

1O Invalid (8) L4

F

$$\begin{aligned}4.3.1 \text{ Grade 11 boys} &= 191 - (54 + 73) \quad \text{OR} \quad 137 - 73 \\ &= 191 - 127 \blacksquare = 64 \\ &= 64 \blacksquare\end{aligned}$$

$$\begin{aligned}\text{Total number of learners} &= 158 + 137 + 99 \\ &= 394 \blacksquare\end{aligned}$$

$$\% = 64$$

$$\begin{aligned}394 \times 100 \blacksquare \\ &= 16,2\% \blacksquare\end{aligned}$$

1M Subtracting

1CA No. of boys in Gr.

11

1CA Total number of learners

1M Multiply by 100

1CA Percentage

NPR (5) L3

D

4.3.2 From Grade 10 to Grade 12 the number of learners is decreasing. ■■

Learners failed ■■ OR

Learners dropped out ■■

OR

Accept a ny other relevant reason 2O Trend

2O Reason

(4) L4

D

[27]

TOTAL: 100

NATIONAL
SENIOR CERTIFICATE

GRADE 1 1

NOVEMBER 2017

MATHEMATICAL LITERACY P2
MARKING GUIDELINE

MARKS: 100

Symbol Explanation

M Method

MA Method with accuracy

CA Consistent accuracy

A Accuracy

C Conversion

S Simplification

RT/RG/RM Reading from a table/Reading from a graph/Read ing from a map

F Choosing the correct formula

SF Substitution in a formula

J Justification

P Penalty, e.g. for no units, incorrect rounding off etc.

R Rounding Off/Reason

AO Answer only

NPR No penalty for rounding

This marking guideline consists of 8 pages.

2 MATHEMATICAL LITERACY P2 (EC/NOVEMBER 2017)

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QUESTION 1 [29]

Question Solution Explanation Topic
and
Level

$$\begin{aligned} 1.1 \text{ Total number of cars and minibuses} &= (10 \times 5) + (5 \times 5) \\ &= 50 + 25 \\ &= 75 \blacksquare \end{aligned}$$

$$\begin{aligned} \text{Probability} &= \frac{50}{75} \blacksquare \end{aligned}$$

Accept 0,667 OR 66,7% 1 M Calculated total
number of cars and
minibuses

1 A Numerator
1A Denominator

(3) P
L2

$$\begin{aligned} 1.2 \text{ Amount for cars washed Monday – Friday:} \\ 10 \times 30 &= 300 \times 5 \blacksquare \\ &= R1\,500 \blacksquare \end{aligned}$$

$$\begin{aligned} \text{Amount for cars Saturday and Sunday} &= 160 \\ 100 \blacksquare \times 10 & \\ &= 16 \blacksquare \\ &= 16 \times 30 \times 2 \\ &= R960 \blacksquare \end{aligned}$$

$$\begin{aligned} \text{Amount for minibuses (Monday – Friday)} \\ &= 5 \times 40 \\ &= R200 \\ 200 \times 5 & \\ &= R1\,000 \blacksquare \end{aligned}$$

$$\begin{aligned} \text{Amount for minibuses (Saturday and Sunday):} \\ 100 \times 5 &= 500 \\ 100 \times 5 &= 500 \times 7 \blacksquare \\ &= 3\,500 \\ 7 \times 40 \times 2 & \\ &= R560 \blacksquare \end{aligned}$$

$$\begin{aligned} \text{Total Amount} &= 1\,500 + 1\,000 + 960 + 560 \\ &= R4\,020 \blacksquare \end{aligned}$$

Claim is valid $\blacksquare$ 1M Multiplied by 10
and 5
1CA Amount for cars

1M Increase by 60%
1CA Number of cars
1CA Amount for cars
on Saturday and
Sunday

1CA Number of
minibuses

1CA Number of

microbuses
1CA Amount for
microbuses

1CA Total amount
1 O Claim valid

(10) F
L4

(EC/NOVEMBER 2017) MATHEMATICAL LITERACY P2 3
Copyright reserved Please turn over
1.3 Water per car = 25×3
= 75 litres

Water for cars in 5 days = $10 \times 5 \times 75$
= 3 750 litres ■

Water for cars on Saturdays and Sundays = $16 \times 2 \times 75$
= 2 400 litres ■

Water per microbus = 25×4
= 100 litres

Water per microbus in 5 days = $5 \times 5 \times 100$
= 2 500 litres ■

Water per microbus on Saturdays and Sundays
= $7 \times 2 \times 100$
= 1 400 litres ■

Total number of litres = $3\,750 + 2\,400 + 2\,500 + 1\,400$ ■
= 10 050 litres ■

OR
Amount of water
= amount for cars + amount for microbuses
= $(3 \times 25 \times 82)$ ■ + $(4 \times 25 \times 39)$ ■■
= 6 150 litres ■ + 3 900 litres ■
= 10 050 litres ■ CA from 1.1

1M Water for cars
Mondays to Fridays

1CA Water for cars on
Saturday and Sunday

1CA Water for
microbuses Mondays to
Fridays

1CA Microbuses on
Saturdays and Sundays

1M Addition
1CA Answer

(6) M

L3

1.4 Water for 7 days = 1 0 050 litres
Water for a month = $1\ 0\ 050 \times 4$
= 40 2 00 litres ■
40 200 l = 40,2 kilolitres ■

Cost of water from table:

$6 \times 8.66 = R51,96$ ■
 $9 \times 10.02 = R90,18$ ■
 $15 \times 12,28 = R184,20$ ■
 $10,2 \times 15.25 = R155,55$ ■

Total amount = R481 ,89
= $R481,89 \times 1,14$ ■
= R549 ,35 ■ CA from 1.3

1MA Amount of litres
1C Amount of kilolitres

1M First tariff
1M Second tariff
1M Third tariff
1M Fourth tariff

1M Calculated VAT

1CA Amount with
VAT

(8) M&F

L3

1.5 Most people do not work on Saturdays and Sundays. ■■
It is the only time they have available to wash their
cars. ■■

Accept any other relevant answer . 2O Opinion

(2) DH

L4

4 MATHEMATICAL LITERACY P2 (EC/NOVEMBER 2017)

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QUESTION 2 [24]

2.1 2.1.1 R212 ■■

1RT Correct value from

table

(2) DH

L2

2.1.2 $188 - 76$ ■

76×100

$= 147,37\%$ ■OR $147,4\%$

Statement is valid ■

% increase is more than 100% 1M Used correct values

1M Divided by correct

value

1CA %

1J Valid

NPR

(4) F

L4

2.1.3 Cost of burying a 5-year-old is less than that of
burying a 15-year old. ■■

Accept any other relevant answers .

2J Opinion

(2) F

L4

2.2 2.2.1 Diameter = 15 0 cm

Length of figure = 100 cm

$A = 100 - 60$ ■

$$= 40 \text{ cm} \blacksquare$$

$$B = 150 - 100$$

$$= 50 \text{ cm} \div 2 \blacksquare$$

$$= 25 \text{ cm} \blacksquare$$

$$25 : 40 \blacksquare$$

$$= 5 : 8 \quad 1M \text{ Subtraction}$$

1M Used 60

1CA Value of A

1 Divided by 2

1CA Answer

1M Ratio

No penalty if not simplified

Penalise 1 mark if ratio

is written with units

(6) M

L4

(EC/NOVEMBER 2017) MATHEMATICAL LITERACY P2 5

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2.2.2 Area of cross shaped glass = Area of longer rectangle

+ 2 × area of smaller rectangles

$$= l \times b + 2(l \times b)$$

$$= 100 \times 40 + 2(30 \times 40) \blacksquare$$

$$= 4\,000 + 2\,400$$

$$= 6\,400 \text{ cm}^2 \blacksquare$$

OR

$$4(40 \times 30) + (40 \times 40) \blacksquare$$

$$= 6\,400 \text{ cm}^2 \blacksquare$$

OR

$$2(100 \times 40) - 402$$

$$= 8\,000 - 1\,600 \blacksquare$$

$$= 6\,400 \text{ cm}^2 \blacksquare$$

$$6\,400$$

$$10\,000 = 0,64 \text{ m}^2 \blacksquare$$

Area wasted = Area of circle – Area of shape

$$= \pi r^2 - 0,64$$

$$= 3,142 \times 0,75 \times 0,75 \blacksquare - 0,64$$

$$= 1,767375 - 0,64 \blacksquare$$

$$= 1,127375 \text{ m}^2 \blacksquare$$

$$\text{Amount lost} = 1,127375 \times 15$$

$$= R16,91 \blacksquare$$

With VAT = 114

$$100 \times 16,91 \blacksquare$$

$$= R19,28 \blacksquare$$

Statement is valid . $\blacksquare$

1M Substituted in formula

1CA Simplification

1C Convert ed to square
metres (m²)

1M Used formula
1SF Substitution

1 CA Unused area

1CA Total amount
1M Added VAT

1CA Answer

1O Opinion (10) M&F
L4

6 MATHEMATICAL LITERACY P2 (EC/NOVEMBER 2017)
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QUESTION 3 [16]

3.1 Company A:

Mean = 29 000 +25 000 +24 000 +15 000 +15 000 +8 000
6■■■

= 116 000
6

= R19 333,33 ■

Company B:

$$\begin{aligned}\text{Mean} &= \frac{31\,000 + 29\,000 + 17\,000 + 14\,000 + 13\,000 + 7\,000}{6} \\ &= 111\,000 \\ &= R\,18\,500 \blacksquare\end{aligned}$$

$$\begin{aligned}\text{Difference} &= 19\,333,33 - 18\,500 \\ &= R\,833,33 \blacksquare\end{aligned}$$

Statement is not valid

1M Addition
1M Divided by 6

1CA Mean

1CA Mean

1CA Difference
1O Not valid (6) DH
L3&4

3.2 No modal salary for company B. $\blacksquare\blacksquare$ 2A
(2) DH
L2

$$\begin{aligned}\text{3.3 Modal value} &= 15\,000 \blacksquare \\ &15 \\ &100 \times 15\,000 \blacksquare \\ &= R\,2\,250 \blacksquare\end{aligned}$$

$$\begin{aligned}\text{Year 1} &= 105 \\ &100 \times 2\,250 \blacksquare \\ &= R\,2\,362,50 \blacksquare\end{aligned}$$

$$\begin{aligned}\text{Year 2} &= 105 \\ &100 \times 2\,362,50 \\ &= R\,2\,480,63 \blacksquare\end{aligned}$$

1RT Correct modal value
1M Calculated 15% of modal value
1CA Value

1M Increased by

5%
1CA First year

1CA Final amount (6) DH&F
L3

3.4 0 OR 0% OR Impossible ■■
2A Answer (2) DH
L2

(EC/NOVEMBER 2017) MATHEMATICAL LITERACY P2 7
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QUESTION 4 [31]

4.1 4.1.1 $A = 21\,168\,700 + 2\,305\,800 + 677\,000 + 2\,214\,400$
 $= 26\,365\,900$ ■

$B = 22\,165\,000$
 $27\,635\,900 \div 100$
 $= 80,2\%$ ■ 1 MA Added correct
values
1CA Total

1 Correct values
1CA % (4) DH
L2

4.1.2 Indian / Asian ■■ 2A Answer
(2) DH
L2

4.1.3 Two million two hundred and fourteen thousand four
hundred ■■ 2A Answer
(2) DH
L2

4.2 4.2.1 $57 - 38 = 19$ ■ 1M Correct number of
seats on the left
1M Subtraction
1M Correct number of
seats on the right
1CA Answer (4) M&P
L2

4.2.2 Percentage of seats for the handicapped 4
 38×100 ■■

10,526315789

10,53% ■ 1RM Correct values
numerator and
denominator
1M Multiplied by 100
1CA Answer (3) M&P
L3

4.3 4.3.1 Total distance $= 35 + 57 + 11 + 21 + 59 + 41$ ■
 $= 224$ km ■

Speed = ■■■■■■■■
■■■■■

$105 = 224$

■■■■■ ■

Time $= \frac{224}{105}$

$= 2,133$ hours ■

Minutes $= 0,133 \times 60$ ■
 $= 7,98$

2 hours 8 minutes ■

Time of arrival = 7 :00 + 2 hours 8 minutes
= 9:08 am ■

Claim is valid ■ 1RM Correct values

1CA Distance

1M Substituting in
formula

1 S Change subject of
formula

1CA Time in hours

1C Time in minutes

1CA Arrival time

1O Claim valid (8) M,
M&P
L3&4

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4.3.2

Amount before increase = 2,82

1,068 ■

= R2,64 ■

Total distance travelled = 224×2
= 448 km ■

Amount = $448 \times 2,64$ ■

= R1 182 ,72 ■

= R1 183 ■ CA from 4.3.1

1M Divided by 1,068

1A Amount

1 CA Distance

1M Multiplied by
amount calculated

1CA Amount

1R Nearest rand (6) F
L3

4.3.3 R61 ■
R349 ■ 1A R61
1A R349 (2) M&P
L2

TOTAL: 100

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GRADE 11

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MATHEMATICAL LITERACY P2
MEMORANDUM

MARKS: 100

Symbol Explanation

M Method

MA Method with accuracy

CA Consistent accuracy

A Accuracy

C Conversion

S Simplification

RT/RG/RM Reading from a table/Reading from a graph/Read from map

F Choosing the correct formula

SF Substitution in a formula

J Justification

P Penalty, e.g. for no units, incorrect rounding off etc.

R Rounding Off/Reason

O Opinion

This memorandum consists of 8 pages.

2 MATHEMATICAL LITERACY P2 (EC/NOVEMBER 2016)

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QUESTION 1 [23]

Ques. Solution Explanation Level

1.1.1 Electricity ■■

Water ■■

Transport ■■

Accept any relevant answers . 2A One expense
excluding cost of
ingredients
(2) L2

1.1.2 Costs for 24 muffins

Muffin mix 500 g for 12 muffins

1 kg of muffin mix = R12,00

500 g = $\frac{1}{2}$ kg = R6 ,00

$$\blacksquare \text{ For 24 muffins} = R6,00 \times 2 \\ = R12 \blacksquare$$

$$\text{Eggs: } 6 = R10,50 \\ 2 = 10,50$$

$$6 \times 2 \\ = R3,50 \blacksquare$$

$$\text{Oil } \frac{1}{2} \text{ cup} = 250 \text{ m} \blacksquare \\ 750 \text{ m} \blacksquare = R12,50$$

$$\blacksquare 12,50 \\ 750 \times 250 \\ = R4,17 \blacksquare 1M$$

$$\text{Milk 3 cups} = 750 \text{ m} \blacksquare \\ 1\ 000 \text{ m} \blacksquare = R11,50$$

$$\blacksquare 750 \text{ m} \blacksquare = 11,50 \\ 1\ 000 \times 750 \\ = R8,63 \blacksquare 1M$$

$$\text{Total amount} = R12,00 + R3,50 + R4,17 + R8,63 \\ = R28,30 \blacksquare$$

OR

$$\text{Muffin mix} = 12,00 \\ 1\ 000 \times 500 \\ = R6,00 \blacksquare$$

$$\text{Eggs} = 10,50 \\ 6$$

$$= R1,75 \blacksquare$$

$$\text{Oil} = 12,50 \\ 750 \times 125 \\ = R2,08 \checkmark$$

$$\text{Milk} = 11,50 \\ 1\ 000 \times 375 \\ = R4,31 \checkmark$$

$$\text{Total} = 6 + 1,75 + 2,08 + 4,31 \\ = 14,14 \times 2 \\ = R28,28 \checkmark$$

1M For muffin mix

1M for eggs

1M for oil

1M for milk

1CA Total amount

(5) L3

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1.1.3 Cost R 28,30 for 24 muffins

■ For 1 200 = 28,30

24 × 1 200

= R1 415 ■

Selling price = R30 per dozen

1 200 = 1 200 / 12

= 100 dozen ■

Money received for 120 dozen = 100 × 30
= R3 000 ■

Profit = R3 000 – R1 415

= R1 585 ■

Statement is valid ■

OR

28,28

2 = 14,14 × 100 = R1 414 (e xpenses) ■

Income = 100 × R30 ■

= R3 000 ■

Profit = R3 000 – R1 414

= R1 586 ■

Statement is valid . ■ CA from 1.1 .2

1MA expenses for
1 200 muffins

1CA number of
dozen

1MA Income for
1 200 muffins

1CA Difference

1 O

(5) L3

1.1.4 In 25 minutes she makes 24 muffins

■ 240 muffins = 25

$$24 \times 240$$

$$= 250 \text{ minutes} \blacksquare$$

She will bake 10 times.

Time spent on cleaning pans:

$$10 \times 10 \text{ minutes} = 100 \text{ minutes} \blacksquare$$

$$\text{Total time} = 250 + 100 = 350 \text{ minutes}$$

$$\text{Time in hours} = 350 / 60$$

$$= 5,833333333 \text{ hours}$$

$$= 5 \text{ hours } 50 \text{ minutes} \blacksquare$$

Time taken to reach destination:

$$\text{Time} = 50 / 100$$

$$= 0,5$$

$$= 30 \text{ minutes} \blacksquare$$

$$\text{Total time} = 5 \text{ h}50 + 0\text{h}30$$

$$= 6 \text{ hours } 20 \text{ minutes} \blacksquare$$

$$\text{Time arrived} = 09:00 + 6 \text{ hours } 20 \text{ minutes}$$

$$= 15:20 \blacksquare$$

She will arrive on time .

1MA Time for
baking

1CA time for
cleaning

1CA Convert to
hours and minutes

1C driving time

1CA adding time

1CA time of arrival
and statement
(6) L4

4 MATHEMATICAL LITERACY P2 (EC/NOVEMBER 2016)

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$$1.2 \text{ Cleaner} = R 50 \times 5 \times 20$$

$$= 5\,000 \blacksquare$$

$$\text{Driver} = (175$$

$$100 \times 50)$$

$$\text{Driver per month} = R 87,50 \blacksquare$$

$$= R87,50 \times 5 \times 20$$

$$= 8\,750 \blacksquare$$

$$\text{Assistant} = 50 + 25$$

$$= 75$$

$$\text{Assistant per month} = 75 \times 5 \times 20$$

$$= 7\,500 \blacksquare$$

$$\text{Total spent on workers} = 5\,000 + 8\,750 + 7\,500$$

$$= R21\,250 \blacksquare \text{ 1M Salary for cleaner}$$

1M calculating 75% more

1CA Salary for driver per month

1M helper per month

1CA total
(5) L3
[23]

QUESTION 2 [27]

Ques. Solution Explanation Level

2.1.1

■

5 16 16 21 22 34 34 35 35 35 42 42 42 42
30

$100 \times 50 = 15$ ■

13 learners were able to get more than 30% and only one learner got less than 30% therefore the learners performance was good . ■■

2M Calculate the number who passes

20

(4) L4

2.1.2

$5+16+16+21+22+34+34+35+35+35+42+42+42+42$

14 ■ ■

$= 421 / 14$

$= 30,07$ ■ CA from 2.1.1

1M Adding values

1A Divide by 14

1 CA mean

(3) L3

2.1.3 Mode = 42 ■

Median:

5 16 16 21 22 34 34 35 35 35 42 42 42 42

$35+34$ ■

$2 = 34,5$ ■

■ Difference = $42 - 34,5$

$= 7,5$ ■ 1A mode

2MA finding the median

1CA Difference

(4) L4

2.1.4 50

$100 \times 50 = 25$ ■

■ 5

14×100

$= 35,71\% \checkmark$

Accept 35, 7% 1M Calculate value of 50%

1M probability as a

fraction
1CA percentage
(3) L3

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2.2.1 Mathematical Literacy = 50 ■■

Math ematics = 50 ■ 2A First answer

1A Second answer

(3) L2

2.2.2 Math ematics = 83 ■■■ 3A C orrect exam
number

(3) L2

2.2.3 ■ ■

$60 - 54 = 6$ ■

1A for left

1A for right

1CA difference

(3) L2

2.2.4 Ratio = 1 : 30

Number of learners = $(9 \times 6) + (10 \times 10) + (6 \times 10)$ ■

= $54 + 100 + 60$

■ = 214 ■

$214 / 30 = 7,1$

■ 8 invigilators will be needed ■

1M Total number

1CA Total

1CA Dividing by 30

1R

(4) L4

[27]

QUESTION 3 [21]

Ques. Solution Explanation Level

3.1.1

Subjects Learner A

APS score Learner B

APS score

IsiXhosa 6 5

English FAL 5 5

Math ematics 4 5

Life Orientation 3 3

Accounting 4 3

Economics 3 5

Business Studies 6 5

Total: 31 31

Both learners have the same APS score . ■ O 1A

1A For each
2 correct
for both
learners

1A

1A LO

1CA Adding all
values
1O Conclusion
(6) L4

6 MATHEMATICAL LITERACY P2 (EC/NOVEMBER 2016)
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3.1.2
CA from
3.1.1

1A for each
two
compound
bars

(5) L2

3.1.3 Equal chances ■■ 2O
(2) L2

3.2.1 Rent = R1 650 × 11
= R18 150 ■
Food = 1 500 × 10
= R15 000 ■

Total amount = R18 150 + R15 000 ■
= 33 150 ■ 1M

Calculating
rent
1M
Calculating
food
1M Addition
1CA Total
(4) L2

3.2.2 ■■
Share room with someone OR to save on costs OR no privacy .

2O
Explanation
(2) L4
[19]

01020304050607080Percentage
SubjectsPercentages of 2 Grade 12 learners
Learner A
Learner B
(EC/NOVEMBER 2016) MATHEMATICAL LITERACY P2 7
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QUESTION 4 [31]

Ques. Solution Explanation Level
4.1 Triangular window:
Base = 150 cm = 1, 5 m
Height = 100 cm = 1 m ■
■ Area of triangular window = $\frac{1}{2}$ base × height

$$= \frac{1}{2} \times 1,5 \times 1 \blacksquare$$

$$= 0,75 \text{ m}^2$$

$$\text{Total} = 0,75 \times 4$$

$$= 3 \text{ m}^2 \blacksquare$$

Circular window:

$$\text{Radius} = \frac{1}{2} \times 1,5$$

$$= 0,75 \text{ m} \blacksquare$$

$$\blacksquare \text{ Area of circular window} = \blacksquare \times r^2$$

$$= 3,142 \times 0,75 \times 0,75 \blacksquare$$

$$= 1,767375 \text{ m}^2$$

$$\text{Total} = 1,767375 \times 4$$

$$= 7,069 \text{ m}^2 \blacksquare$$

Square window:

$$\text{Width} = 1,5 \text{ m}$$

$$\text{Length} = 1,5 \text{ m}$$

$$\blacksquare \text{ Area of square window} = S^2$$

$$= 1,5 \times 1,5$$

$$= 2,25 \text{ m}^2$$

$$\text{Total} = 2,25 \times 4$$

$$= 9 \text{ m}^2 \blacksquare$$

$$\text{Total area of windows for all four walls} = 3 + 7,069 + 9 \blacksquare$$

$$= 19,069 \text{ m}^2 \blacksquare$$

$$\text{Area of wall of 1 classroom} = l \times b$$

$$= 12 \times 5$$

$$= 60 \text{ m}^2 \blacksquare$$

$$\text{Total area for four classrooms} = 60 \times 4 \blacksquare$$

$$= 240 \text{ m}^2 \blacksquare$$

Area of wall to be painted = total area of walls – total area of windows

$$= 240 - 19,069 \blacksquare$$

$$= 220,9305 \text{ m}^2$$

$$= 221 \text{ m}^2 \blacksquare \text{ 1C covert to m}$$

1SF

1CA answer

1A radius

1SF

1CA area of four circular window

1CA area of four square windows

1CA adding the areas

1CA total of areas

1M area of 1 classroom wall

1M for $\times 4$ area
of 4 classrooms

1CA answer

1CA subtraction

1CA Wall area
to be painted
(14) L3

8 MATHEMATICAL LITERACY P2 (EC/NOVEMBER 2016)

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4.2 Litres required for the wall s:

$$8 \text{ m}^2 = 1 \blacksquare$$

$$221 \text{ m}^2 = 1 / 1 \times 221 / 8 \\ = 27,625 \text{ litres } \blacksquare$$

If 5 litre tins are bought:

$$27,625 / 5 = 5,525 = 6 \text{ tins of paint } \blacksquare (5 \blacksquare)$$

$$\text{Cost} = 6 \times 105 \blacksquare \\ = \text{R}630$$

If 20 litres tins are bought:

$$27,625 / 20 = 1,38 = 2 \blacksquare$$

$$\text{Cost} = 2 \times 405 \\ = \text{R}810 \blacksquare$$

Their statement is valid . $\blacksquare$ CA from 4.1.1

1MA

1CA

1M $\times$ by 105

1M

1CA

1O
(6) L4

4.3.1 5 $\blacksquare$ tins:

$$\text{Radius} = 9 \text{ cm}$$

$$\text{Diameter} = 18 \text{ cm}$$

Box A:

Number of t ins over the length of the box:

$$130/18 = 7 \blacksquare$$

Number of tins over the height of the box:

$$25/24 = 1,04 = 1 \blacksquare$$

Number of t ins over the width of the box:

$$104/18 = 5,77 = 5 \blacksquare$$

$$\text{Total tins for Box A} = 7 \times 1 \times 5 \\ = 35 \text{ tins } \blacksquare$$

Box B:

$$l = 65 \text{ cm}$$

$$b = 49 \text{ cm}$$

$$h = 52 \text{ cm}$$

Number of tins on the width of the box:

$$65/18 = 3, 6 = 3$$

Number tins along the length of the box:

$$49/24 = 2,04 = 2$$

$$\text{No. of tins on height} = 52 / 18 = 2,88 = 2$$

$$\begin{aligned} \text{Total number of tins in box B} &= 3 \times 2 \times 2 \\ &= 12 \text{ tins} \blacksquare \end{aligned}$$

Box A can carry 28 tins which is more than 2 times 12 so their argument is valid . $\blacksquare$

1M number of tins
over length

1M number of tins
over width

1M number of boxes
over height

1CA total number of
tins in Box A

1CA total number of
boxes in Box B

1O

(6) L3

$$4.3.2 \text{ Tins in Box B} = 12 \blacksquare$$

$$\begin{aligned} \text{Tins in 20 boxes} &= 12 \times 20 \\ &= 240 \blacksquare \end{aligned}$$

$\blacksquare$

$$\begin{aligned} \text{Amount lost} &= R105 \times 240 \blacksquare \\ &= R25\,200 \blacksquare \text{ 1CA from 4.3.1} \end{aligned}$$

1CA number of tins

1M $\times$ by 240

1A Price of paint

1CA Total amount

(5) L3

[31]

TOTAL: 100

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GRADE 11

NOVEMBER 2015

MATHEMATICAL LITERACY P2
MEMORANDUM

MARKS: 100

Symbol Explanation

M Method

MA Method with accuracy

CA Consistent accuracy

A Accuracy

C Conversion

S Simplification

RT/RG/RM Reading from a table/Reading from a graph/Read from map

F Choosing the correct formula

SF Substitution in a formula

J Justification

P Penalty, e.g. for no units, incorrect rounding off, etc.

R Rounding Off/Reason

This memorandum consists of 6 pages.

2 MATHEMATICAL LITERACY P2 (EC/NOVEMBER 2015)

Copyright reserved Please turn over QUESTION 1

1.1 1.1.1 5 Entrances ■■ 2A Identifying

the number of

entrances (2)

1.1.2 ■ Walk straight towards Shop No. 6 and turn left. ■

■ Edgars will be on your left continue until you get to Shop No. 60 and turn right. ■

■ Keep walking until Shop No. 52 -53 and turn left where Entrance 2 is ■

Accept any reasonable explanation 3A

Explanation

(3)

1.2 1.2.1 1h ■57 minutes ■ 1A Hours

1A Minutes (2)

1.2.2 11:20 + 45 min + 15 min ■ = 12:20 ■

It will be too late to watch the 12:15 screening ■

$14:45 + 1:57 = 16:42 + 40 \text{ min} \blacksquare = 17:22 \text{ min} \blacksquare$

The only time slot will be the 14:45 screening and still be at home on time. ■ 1M Adding

45 min and

15 min

1A

1O Opinion

1A Adding

1h57 and

40 min

1A Time to

arrive at home

1O

Conclusion (6)

1.2.3 ■ Cleaning of the cinema ■

■ To prepare for the next showing ■

OR

■ Allow the cinema crew to take a break ■

Accept any other relevant reasons 1A First

reason

1A Second

reason

(2)

1.3 1.3.1 Regular pricing = $75 + 55 \blacksquare\blacksquare = R130 \blacksquare$ 1A Identifying the correct

values

1M Adding

1A (3)

1.3.2 $\blacksquare$ It is too expensive on a Sunday or weekends $\blacksquare\blacksquare$

OR

$\blacksquare$ On other days its much cheaper $\blacksquare\blacksquare$

Accept any other explanation 2O

Explanation

(2)

[20]

(EC/NOVEMBER 2015) MATHEMATICAL LITERACY P2 3

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2.1 2.1.1 (a) Total cost (in rand) = $R200 \blacksquare + \text{number of minutes more than } 150 \blacksquare \times R0,60 \blacksquare$ 1A Rental

1A Minutes

more than 150

1A Multiply by

60 cents

OR

Total cost (in rand) = $R 200 \blacksquare + (\text{number of minutes more than } 150) \blacksquare \times R0,60 \blacksquare$ 1A Rental

1A Minutes

more than 150

1A Multiply by

60 cents

OR

Total cost (in rand) = $R200 \blacksquare + (\text{number of minutes} - 150) \blacksquare \times R0,60 \blacksquare$ 1A Rental

1A Minutes

more than 150

1A Multiply by

60 cents (3)

(b) Total cost (in rand) = $R200 + (\text{number of minutes} - 150) \times R0,60$

$= 200 + (625 - 150) \times 0,60 \blacksquare$

$= 200 + (475 \blacksquare \times 0,60)$

$= 200 + 285 \blacksquare$

$= R485,00 \blacksquare$ CA from

2.1.1 (a)

1SF

1S

1S

1A (4)

2.1.2

1 Mark for line

0–100

minutes

1 Mark for the

correct break -

even point

2 Marks for

any other 3

points plotted

correctly

1M Labelling

Call Package

A (5)

2.1.3 For the same number of minutes used the same amount will be paid for both Call packages 1A refer to minutes

1A refer to

cost (2)

2.1.4 Break -even point (450 ;380) CA from 2.1.2

1CA 450

minutes

1CA 380 rand

If order

incorrect 0 (2)

Call package A

Call package B

4 MATHEMATICAL LITERACY P2 (EC/NOVEMBER 2015)

Copyright reserved Please turn over 2.1.5 Call package A

Total cost = $100 + (451 - 100) \times 0,80$

= $100 + (351 \times 0,80)$

= $100 + 280,8$

= R 380,80

Call package B

Total cost = $200 + (451 - 150) \times 0,60$

= $200 + (301 \times 0,60)$

= $200 + 180,60$

= R 380,60

Difference = R 380,80 – R 380,60

= R 0,20

OR

Call package A - Call package B

= $100 + (451 - 100) \times 0,80 - 200 + (451 - 150) \times 0,60$

= $100 + (351 \times 0,80) - 200 + (301 \times 0,60)$

= $100 + 280,80 - 200 + 180,60$

= R 380,80 – R 380,60

= R 0,20

1SF

1CA

1CA

1CA

Difference

1SF

1CA

1CA

1CA

Difference (4)

2.1.6 Would recommend that Mr. Gardner use Call

Package A ■

With R 300 on Call Package A he will get 350 minutes ■■

With R 300 on Call Package B he will get 316 minutes to 317 minutes ■■ 1CA Choosing

correct
package

2 CA

2 A (5)

2.2 Number of households = $10 \times 10 \times 10 \times 10$ ■
= 10 000 ■ 1A Identifying

10

1M Multiplying

10 four times

1CA (3)

[28]

(EC/NOVEMBER 2015) MATHEMATICAL LITERACY P2 5

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3.1 3.1.1 $A = 17,20 \times 1,14$ ■

= R 19,61 ■

B = 31,95 ■

1,14

= R 28,03 ■ OR $A = 17,20 + (17,20 \times 0,14)$ ■

= $17,20 + 2,41$

= R 19,61 ■ 1M

1A

1M

1A (4)

3.1.2 No■, she will not pay for the first 6 kl ■ 1A

1O (2)

3.1.3 Payment for 19,5 kl

= $(6 \times 0) + (4,5 \times 7,60) + (9 \times 11,61)$ ■

= R 0 + R 34,20 + R 104,49 ■

= R 138,69 ■ $\times 1,14$

= R 158,11 ■ 1M

1S

1A

1A Including

VAT (4)

3.2 3.2.1 Diameter = 750 mm,
therefore radius = 375 mm ■ = 0,375 m

Height = 2230 mm = 2,230 m ■

Volume = $3,142 \times 0,375^2 \times 2,230$ m ■

= $3,142 \times 0,140625 \times 2,230$ m

= $0,985311562 \text{ m}^3$ ■ $\times 1000$

= 985,311562 /

Accept 985,312 ■

Therefore;

$1000 \text{ l} \neq 985,311562 \text{ l}$ ■ OR 985,312 ■ 1A Finding
radius

1A Convert

mm to m

(both)

1SF

1S radius²

1CA in litres

1A (6)

3.2.2 The given volume was rounded to the nearest

1000 ■

Consumers are under the impression that you buy a tank with a capacity of 1 000 l, while it only has a tank capacity of 985 l. ■■ 1O

2O (3)

3.2.3 Tank can overflow if it is filled above the recommended filled height ■■ 2R (2)

3.2.4 Any of the vertical round water tanks ■■ 2A (2)

3.3 Bar graph ■and

Pie chart ■ 1A Bar graph

1A Pie chart (2)

[25]

6 MATHEMATICAL LITERACY P2 (EC/NOVEMBER 2015)

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QUESTION 4

4.1 4.1.1 Mean = $\frac{26 + 29 + 22 + 23 + 22 + 21 + 24}{7}$ ■

= 167

7

= 23,85714286 ■

= 24 °C ■ 1M adding all

day temp

1M/7

1CA

1CA (4)

4.1.2 Range = $29^{\circ}\text{C} - 15^{\circ}\text{C}$ ■
= 14°C ■ 1M concept of

range

1A (2)

4.1.3 Median = 15; 16; 17; 17; 17; 20; 21 ■
= 17°C ■ 1M Arrange

values

1A (2)

4.1.4 Night temperatures decreases from Wednesday to Friday and then increases on Saturday and then from Sunday remains constant until Tuesday ■■ 2O (2)

4.1.5 Monday ■with

4°C ■ 1RT Day

1A Difference (2)

4.1.6 Friday ■■ 2RT (2)

4.1.7

Any 4 pairs

correctly

plotted

1Mark for

legend

1 Mark correct

graph (6)

4.2 It is highly unlikely that it will rain on Friday because it is only 30% which is small chance for rain to fall ■■ 2O (2)

4.3 4.3.1 Time = Distance

Speed

= $64,7 \text{ km}$ ■

90 km/h
= 0,7188....hours x 60 ■
= 43,133...
= 43 minutes ■

1M

1S

1A (3)

4.3.2 Your speed will decrease ■ therefore it will take you longer ■ to complete the trip 1A refer to speed

1A refer to

time (2)

[27]

TOTAL: 100

010203040

Wo Do VrSaSoMa DiTemperature in °C

Days Day and night

temperatures for George

from 14 Jan - 20 Jan

2015

Day temp

Night temp

NATIONAL
SENIOR CERTIFICATE

GRADE 1 1

NOVEMBER 2015

MATHEMATICAL LITERACY P1
MEMORANDUM

MARKS: 100

Symbol Explanation

M Method

A Accuracy

CA Consistent accuracy

RT/RG/RM Reading from a table/Reading from a graph/Read from map

RP Reading from the plan

SF Substitution in a formula

S Simplifications

P Penalty (no units, incorrect rounding off etc.)

O Opinion

J Justification

RO Rounding Off

This memorandum consists of 7 pages .

2 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2015)

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QUESTION 1

Question Solution Explanation Mark

1.1 1.1.1 $R169 \times 24 \blacksquare + R55 \times 24 \blacksquare$
 $R4\ 056 + R1\ 320 \blacksquare$
 $= R5\ 376 \blacksquare$ 2M
 1S
 1A (4)

1.1.2 $R0,79 \times 8 \blacksquare$
 $R6,32 \blacksquare$ 1M
 1S (2)

1.1.3 $R1\ 999 + R1\ 999 \times 4,4\% \blacksquare$ OR $R1\ 999 \times 1,044$
 $= R1\ 999 + R87,96 \blacksquare = R2\ 086,96$
 $= R2\ 086,96 \blacksquare$
 OR
 $R1,999 \times 0,144$
 $R2\ 086,86 \blacksquare$ 1M

1S
 (2)

1.2 1.2.1 $R0,35 \times 25 \blacksquare$
 $= R8,75 \blacksquare$ 1M
 1A (2)

1.2.2 $18:15 \blacksquare$
 $= 6:5 \blacksquare$ 1M
 1A (2)

1.2.3 $25 \blacksquare = 5 \blacksquare$
 $\frac{35}{7}$
 $= 0,71$ or 71% (Answer only – full marks) 1M
 1A
 (2)

1.3 1.3.1 Month 1: $R1\ 525 \times 20\%$
 $= R305,00 \blacksquare$
 Month 2: $R1\ 830,00 \times 20\%$
 $= R366,00$
 Interest $= R366 + R305 = R671,00 \blacksquare$
 1M

1S (2)

1.3.2 $R400 \times 11 \times 8 \blacksquare + R2\ 000 \times 8 + R671 \blacksquare$
 $= R35\ 200 + 16\ 000 + R671$
 $= R51\ 871,00 \blacksquare$ 2M
 1CA
 Multiplication
 by different
 number not 11 (3)

1.3.3 $R400 \times 11 + R2000 + R671 \blacksquare = 400\ 000$
 $R4\ 400 + R2\ 000 + R671 \blacksquare$
 $= R7\ 071,00 \blacksquare$
 OR

$+ R671 = R7\ 071,00 \blacksquare$ 1M

1S
 1S
 (3)

Question Solution Explanation Mark

1.4 1.4.1 R4 760 x 11,82 ■ OR 4,760 x 11,82
= R56 263,20 ■ = 56,26 1M
1A (2)

1.4.2 R1 571,05 x 114% ■ OF R1 571,05 x 14%
R1 790,997 R219,95 ■

R1 790,997 x 2 ■ R1 571,05 + R219,95 ■
= R3 581,99 ■ R1 571,05 + R219,95 ■

OR

R1 571,05 x 14%

= R 219,95 ■

= R1 571,05 + R219,95 ■

= R1 790,997 x 2

= R3 581,99 ■

OR

R1 571,05 x 2 ■

= R3 142,10 x 114% ■

= R3 581,99 ■ 1M

1S

1A

(3)

[28]

4 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2015)

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QUESTION 2

Question Solution Explanation Marks

2.1 2.1.1 227 ■■ RD (2)

2.1.2 $7 \div 60$ ■ = 0,1166666667 h

$5 \div (60 \times 60)$ = 0,00138888888 h ■

4 h + 0,1166666667 h + 0,00138888888 h

= 4,118055556 hours

= 4 hours ■ 2M

1S

(3)

2.1.3 Average speed =

■

= 16,5 ■ km/hour ■ 1S

2S

Penalise for unit (3)

2.2 2.2.1 Arrival time – 18:56 ■

Departure time – 19:00 ■ 2RT

(2)

2.2.2 Bloemfontein ■■ 2RT (2)

2.2.3 Area of a bigger circle – Area of a smaller circle ■

$3,142 \times (5 \text{ m})^2 - 3,142 \times (2,5 \text{ m})^2$ ■

= 78,55 m² – 19,6375 m² ■

= 58,91 m² ■ (Accept 58,91 m²) ■ 1F

2SF

1S

1A (5)

[17]

Question Solution Explanation

3.1 3.1.1 $113\,341 \times 26\%$ ■
 $= 113\,341 \times 0,26$
 $= 29\,468,66$
 $= 29\,469$ ■

(2)

3.1.2 (i) Cancer ■
 (ii) Heart disease ■
 (2)

3.1.3
 1 mark per three correctly plotted points = 2 marks
 2 marks for correctly labelling x -axis and y -axis = 2 marks Types of
 death Accidents Cancer Heart
 disease Suicide Homicide All other
 causes
 % death 26 13 12 11 6 32 232

Causes of death ■
 1 mark for completing the table = 1 mark
 1mark for joining points with a straight line = 1 mark
 (6)

3.1.4 $113\,341 - 39\,213$ ■
 $= 74\,128$ ■ 1M
 1A (2)

3.1.5 25–44 years ■■ 2RG (2)

3.1.6 They drive recklessly trying to impress those
 people of their age about their driving skills. They
 do not concentrate on the road trying to greet
 their friends walking on the street. 25 –44 are
 mature enough and they are careful when
 driving . ■■
 (Accept any acceptable reasons .)

2A (2) 05101520253035
 Accidents Cancer Heart
 diseaseSuicide Homicide All other
 causesPercentages % death
 ■
 ■ ■ ■

6 MATHEMATICAL LITERACY P1 (EC/NOVEMBER 2015)
 Copyright reserved Please turn over
 3.2 3.2.1 $15+14 \times 2 + 13 +12 + 9 \times 5 + 10 \times 3+7 + 5 \times 2$ ■
 16

$= 160$ ■
 16
 $= 10$ ■ 1M

1S

1S (3)

3.2.2 Mode = 36 ■■ 2A (2)

3.2.3 Maluti FET College ■■ 2A (2)
[23]

QUESTION 4

Question Solution Explanation
n Marks

4.1 4.1.1 35 passengers ■■ 2A (2)

4.1.2 Get in and turn right. ■ Walk straight or southwards and at the third row from the back ■ next to the window will be your seat. Accept any relevant explanation. 3
(3)

4.1.3 A7 ■■ 2A (2)

4.2 4.2.1 Strip Map ■■ 2A (2)

4.2.2 $430 - 220$ ■ OR $22 + 19 \times 2 + 14 \times 2 + 17 \times 2 + 10 + 35 + 43$
 $= 210$ km ■ $= 210$ km 1M
1S (2)

4.2.3 N1 or Road 1 ■ and 4 toll gates ■ 2RM (2)

4.2.4 North west ■■ 2RM (2)
[15]

(EC/NOVEMBER 2015) MATHEMATICAL LITERACY P1 7
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QUESTION 5

Question Solution Explanation Marks

5.1 5.1.1 323 litres ■
 $323 \times 1\,000$ ■ = 323 000 ■■ 1RD
1C (3)

5.1.2 Interest rate is the rate ■ at which interest is paid or charged for the use of money . ■
OR
It is percentage of the total amount paid over a period of time. 2A
(2)

5.1.3 $R4\,799,00 - R479,90$ ■
 $= R4\,319,10$ ■ 1M
1S (2)

5.1.4 $R4\,319,10 \times (21,75 \div 100 \div 12)$ ■
 $= R4\,319,10 \times 0,018125$ ■
 $= R78,2836875 \times 24$
 $= R1\,878,81$ ■
OR
 $R4\,319,10 \times (21,75 \div 100)$
 $= R939,40425 \times 2$
 $= R1\,878,81$

OR

$$R4 \ 319,10 \times (21,7 \div 100) \times 2$$

$$= R1 \ 878,81 \ 1M$$

1S

1S

(3)

$$5.2.1 \ 32;35;36;37;38;42;46;46;46;49;58;62;64;66;67$$

$$\text{Median} = 46 \quad \blacksquare \blacksquare \ 2A$$

(2)

$$5.2.2 \ IQR = 62 \blacksquare - 37 \blacksquare$$

$$= 25 \blacksquare \ 2CV$$

1A (3)

$$5.2.3 \ 10 \blacksquare$$

$$15 \blacksquare$$

$$= 2$$

$$3 \blacksquare \ 2M$$

1A (3)

[18]

TOTAAL: 100

NASIONALE
SENIOR SERTIFIKAAT

GRAAD 11

NOVEMBER 2014

WISKUNDIGE GELETTERDHEID V1
MEMORANDUM

PUNTE: 100

Hierdie memorandum bestaan uit 6 bladsye .

2 WISKUNDIGE GELETTERDHEID V1 (NOVEMBER 2014)

Kopiereg voorbehou Blaai om asseblief

VRAAG 1

VR OPLOSSING Finansies

$$1.1.1 \ R70,17 \times 3 + 2 \times R2 \ 367,54 + R28,06 \times 3 + R28,94 \quad \blacksquare \ 1M$$

$$R210,51 + R4 \ 735,08 + R84,18 + R28,94 \quad \blacksquare \ 1S$$

$$R5 \ 058,71 \quad \blacksquare \ 1A \ (3)$$

1.1.2

$$\blacksquare \ 1M$$

$$= R1,40 \blacksquare \ 1 \ A \ (2)$$

1.1.3 Desember $\blacksquare \blacksquare \ 2RT \ (2)$

1.1.4 BTW -nommer 8754896368 ■■ 2RT (2)

1.1.5 Snymasjien RYOBI 43CC ■■ 2RT (2)

1.1.6 Q 000002158 ■■ 2RT (2)

Meting

1.1.7 1 000 m■ = 1 ■

2 00 m■ = x ■ 1M

=

0,2 ■ = x ■

1A (2)

1.1.8 12 x 13 ■
= 156 maande ■ (2)

Data

Hantering

1.2.1 Skool E ■■ 2A (2)

1.2.2 Skool A fooie – Skool B fooie

R14 000 – R12 000 ■ 1M

= R2 000,00 ■ 1 A (2)

[21]

(NOVEMBER 2014) WISKUNDIGE GELETTERDHEID V1 3

Kopiereg voorbehou Blaai om asseblief

VRAAG 2

VR OPLOSSING Finansies

2.1.1 ■

■

2SF

= R 168 947,37 ■

OF

R192 600 x

R23 652,63■

R19 6200 – R23 652,63 ■

= R 168 947,37 ■ 1 A

OF

R192 600: x ■ = 114% : 100 ■

x = R168 947,37 ■

(3)

2.1.2

■ = R13,2 8 ■ 1D

1A (2)

2.2.1 Wins = R7 200,00 – R3 339,14 ■

= R3 860,86 ■ 1SF

1 A (2)

2.2.2 R350,00 x 4 = R1 400,00 ■ OF 1% of R350

R1 400,00 x 1% ■ = R3,50 x 4

$$= R14,00 \text{ ■ maand ■} \quad = R14,00 \text{ 1M}$$

1S
1A (3)

$$2.2.3 \quad \text{---}$$

$$\quad \times 100 \text{ ■}$$

$$\quad =$$

$$\quad \times 100 \text{ ■}$$

$$= 5,56 \text{ of } 5,6\% \text{ ■ 1SF}$$

1S

1CA
(3)

$$2.3.1$$

$$\text{■}$$

$$=$$

$$\text{■}$$

$$= 0,00074 \text{ ■ 1D}$$

1 S
1A
(3)

$$2.3.2 \text{ Paalement verskuldig } =$$

$$+ R21,00 \text{ ■}$$

$$= R5 \ 753,05 \div 12 + R21,00 \text{ ■}$$

$$= R500,42 \text{ ■ 1SF}$$

1S
1CA (3)

$$2.3.3 \text{ R6 879 ■}$$

$$379,83 \text{ ■}$$

$$= R \ 18,1107 \text{ ■}$$

$$= R18,11 \text{ 2D}$$

1A
(3)
[22]

4 WISKUNDIGE GELETTERDHEID V1 (NOVEMBER 2014)
Kopiereg voorbehou Blaai om asseblief
VRAAG 3: ME TING

$$3.1.1 \text{ Oppervlakte } = 162 \text{ m} \times 160 \text{ m} \text{ ■}$$

$$= 25 \ 920 \text{ m}^2 \text{ ■ 1SF}$$

1A (2)

$$3.1.2 \text{ Omtrek } = 2(160 \text{ m} + 162 \text{ m}) \text{ ■}$$

$$= 644 \text{ m} \text{ ■ 1SF}$$

1A (2)

$$3.1.3 \text{ Aantal gatte } = 324 \div 2 - 1 \text{ ■}$$

$$= 162 - 1 \text{ ■}$$

$$= 161 \text{ gatte ■ 1SF}$$

1S
1A (3)

$$3.1.4 \text{ BO } = 2(6,2 \text{ m} \times 3 \text{ m}) + 2(5,1 \times 3 \text{ m}) - (1,71 \text{ m}^2 + 4,875 \text{ m}^2) \text{ ■}$$

$$= 37,2 \text{ m}^2 + 30,6 \text{ m}^2 - 6,585 \text{ m}^2 \text{ ■}$$

$$= 61,215 \text{ m}^2 \text{ ■}$$

1SF
1S
1A (3)

$$3.1.5 \quad 1 \text{ m}^2 = 6 \text{ m}^2$$

$$x = 61,215$$

$$61,215 \div 6 = x$$

$$x = 10,2025 / 5$$

$$= 2,0405$$

Daarom 3 (5) blikke verf 1M

1M

1A (3)

3.2

Penaliseer vir geen eenhede

$$= 34,26 \text{ kg/m}^2$$

Aanvaar 114 kg (33,97 kg/m²) 1CV

1SF

1A

1U (4)

[17]

VRAAG 4

4.1.1 C2 2RM (2)

4.1.2 Indiese oseaa n 2RM (2)

4.1.3 Swaziland en Mosambie k 2RM (2)

$$4.1.4 \quad 6,5 \times 2600\,000 = 16\,900\,000$$

$$16\,900\,000 \div 100\,000$$

$$= 169 \text{ km}$$

1C

1A (3)

4.1.5 .Big Bend

.Lavumisa

.Mkuze

.Phinda 3RM

(3)

(NOVEMBER 2014) WISKUNDIGE GELETTERDHEID V1 5

Kopiereg voorbehou Blaai om asseblief

4.1.6 Kwazulu -Natal 2A (2)

4.1.7 Qumbu

Mount Frere 2A

(2)

[16]

VRAAG 5: DATA HANTERING

5.1.1 Limpopo 2RT (2)

5.1.2 Gauteng : Totale aantal babas

$$60 : 320$$

$$: \dots$$

1: 5 ■
2CV
1A
1R (4)

5.1.3

1 Punt per balke streng akkuraat (4)
(slegs die ontbrekende balke in vraestel) (4) 01020304050607080
Oos-Kaap Limpopo Mpumalanga Noord-Kaap Wes-Kaap Gauteng KwaZulu-Natal Aantal geboortes
Provinsies Aantal geboortes soos op 1 Januarie 2014
6 WISKUNDIGE GELETTERDHEID V1 (NOVEMBER 2014)
Kopiereg voorbehou Blaai om asseblief

5.2.1 ■ ■
■

= 5,45 ■ 1 add
1M

1A (3)

5.2.2 Media an = $\frac{R2,76 + R2,91}{2}$ ■
= (■

= R2,84 ■ 1M

1S

1A
(3)

5.2.3 Modale waarde = R6,50 ■■ 2A (2)

5.2.4
■

■ 1M

1A (2)

5.2.5 Variasiewydte = $R3,36 - R2,16$ ■
= R1,20 ■ 2A
(2)

5.2.6 R1,39 ■■ 2RB (2)
[24]

TOTAAL: 100

NATIONAL
SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2014

MATHEMATICAL LITERACY P1

MEMORANDUM

MARKS: 100

This memorandum consists of 6 pages.

2 MATHEMATICAL LITERACY P1 (NOVEMBER 2014)

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QUESTION 1

Q SOLUTION Finance

1.1.1 $R70,17 \times 3 + 2 \times R2\ 367,54 + R28,06 \times 3 + R28,94$ ■ 1M
 $R210,51 + R4\ 735,08 + R84,18 + R\ 28,94$ ■ 1S
 $R5\ 058,71$ ■ 1A (3)

1.1.2
■ 1M

= R1,40 ■ 1A (2)

1.1.3 December ■■ 2RT (2)

1.1.4 VAT number 8754896368 ■■ 2RT (2)

1.1.5 Brush cutter RYOBI 43CC ■■ 2RT (2)

1.1.6 Q 000002158 ■■ 2RT (2)

Measurement
1.1.7 $1\ 000\ m \blacksquare = 1\ \blacksquare$
 $2\ 00\ m \blacksquare = x$ ■ 1M

=

0,2 ■ = x ■

1A (2)

1.1.8 12×13 ■ 1M
= 156 months ■ 1A (2)

Data Handling

1.2.1 School E ■■ 2A (2)

1.2.2 School A fees – School B fees
 $R14\ 000 - R12\ 000$ ■ 1M
= R2 000,00 ■ 1 A (2)
[21]

(NOVEMBER 2014) MATHEMATICAL LITERACY P1 3

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QUESTION 2

Q SOLUTION Finance

2.1.1 ■
■ 2SF
1 A (3)

$$= R168\,947,37 \blacksquare$$

OR

$$R192\,600 \times$$

$$R23\,652,63 \blacksquare$$

$$R19\,6200 - R23\,652,63 \blacksquare$$

$$= R168\,947,37 \blacksquare$$

OR

$$R192\,600 : \blacksquare \blacksquare = 114\% : 100 \blacksquare$$

$$x = R168\,947,37 \blacksquare$$

2.1.2

$$\blacksquare = R13,28 \blacksquare 1D$$

1A (2)

$$2.2.1 \text{ Profit} = R7\,200,00 - R3\,339,14 \blacksquare$$

$$= R3\,860,86 \blacksquare 1SF$$

1 A (2)

$$2.2.2 \text{ } R350,00 \times 4 = R1\,400,00 \blacksquare \text{ OR } 1\% \text{ of } R350$$

$$R1\,400,00 \times 1\% \blacksquare$$

$$R3,50 \times 4$$

$$= R14,00 \text{ a month } \blacksquare$$

$$= R14,00 \text{ 1M}$$

1S

1A (3)

2.2.3

$$\times 100 \blacksquare$$

$$=$$

$$\times 100 \blacksquare$$

$$= 5,56 \text{ or } 5,6\% \blacksquare 1SF$$

1S

1CA

(3)

2.3.1

$$\blacksquare$$

$$=$$

$$\blacksquare$$

$$= 0,00074 \blacksquare 1D$$

1 S

1A

(3)

2.3.2 Instalment due =

$$+ R21,00 \blacksquare$$

$$= R5\,753,05 \div 12 + R21,00 \blacksquare$$

$$= R479,42 + R21,00$$

$$= R500,42 \blacksquare 1SF$$

1S

1CA

(3)

$$2.3.3 \text{ } R6\,879 \blacksquare$$

$$379,83 \blacksquare$$

$$= R18,1107 \blacksquare$$

$$= R18,11 \text{ 2D}$$

1A

(3)

[22]

4 MATHEMATICAL LITERACY P1 (NOVEMBER 2014)

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QUESTION 3: MEASUREMENT

$$\begin{aligned} 3.1.1 \text{ Area} &= 162 \text{ m} \times 160 \text{ m} \blacksquare \\ &= 25\,920 \text{ m}^2 \blacksquare 1\text{SF} \end{aligned}$$

1A (2)

$$\begin{aligned} 3.1.2 \text{ Perimeter} &= 2(160 \text{ m} + 162 \text{ m}) \blacksquare \\ &= 644 \text{ m} \blacksquare 1\text{SF} \end{aligned}$$

1A (2)

$$\begin{aligned} 3.1.3 \text{ No. of holes} &= 324 \div 2 - 1 \blacksquare \\ &= 162 - 1 \blacksquare \\ &= 161 \text{ holes} \blacksquare 1\text{SF} \end{aligned}$$

1S

1A (3)

$$\begin{aligned} 3.1.4 \text{ SA} &= 2(6,2 \text{ m} \times 3 \text{ m}) + 2(5,1 \times 3 \text{ m}) - (1,71 \text{ m}^2 + 4,875 \text{ m}^2) \blacksquare \\ &= 37,2 \text{ m}^2 + 30,6 \text{ m}^2 - 6,585 \text{ m}^2 \blacksquare \\ &= 61,215 \text{ m}^2 \blacksquare \end{aligned}$$

1SF

1S

1A (3)

$$3.1.5 \quad 1 \blacksquare = 6 \text{ m}^2$$

$$x = 61,215 \blacksquare$$

$$61,215 \div 6 = x$$

$$x = 10,2025 / 5 \blacksquare$$

$$= 2,0405$$

Therefore 3 (5 $\blacksquare$) tins of paint . $\blacksquare$ 1M

1M

1A (3)

3.2 $\blacksquare$

$\blacksquare$ Penalise if units not written

$$= 34,26 \text{ kg/m}^2 \blacksquare \blacksquare$$

Accept 114 kg (33,97 kg/m²) 1CV

1SF

1A

1U (4)

[17]

QUESTION 4

4.1.1 C2 $\blacksquare \blacksquare$ 2RM (2)

4.1.2 Indian Ocean $\blacksquare \blacksquare$ 2RM (2)

4.1.3 Swaziland $\blacksquare$ and Mozambique $\blacksquare$ 2RM (2)

$$4.1.4 \quad 6,5 \times 2 \quad 600\,000 = 16\,900\,000 \blacksquare$$

$$16\,900\,000 \div 100\,000 \blacksquare$$

$$= 169 \text{ km} \blacksquare 1\text{M}$$

1C

1A L3(3)

4.1.5 .Big Bend $\blacksquare$

.Lavumisa ■

.Mkuze ■

.Phinda 3RM

L1(3)

(NOVEMBER 2014) MATHEMATICAL LITERACY P1 5

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4.1.6 Kwazulu -Natal ■■ 2A (2)

4.1.7 Qumbu ■

Mount Frere ■ 2A

(2)

[16]

QUESTION 5: DATA HANDLING

5.1.1 Limpopo ■■ 2RT (2)

5.1.2 Gauteng : Total number of babies

60 : 320 ■■

: . . . ■

1: 5 ■

2CV

1A

1R (4)

5.1.3

1 Mark per bar strictly accurate (4)

(only the missing ones in the paper) (4)

01020304050607080

Eastern Cape Limpopo Mpumalanga Northern Cape Western Cape Gauteng KwaZulu Natal Number of births

Provinces Number of births as on 1 January 2014

6 MATHEMATICAL LITERACY P1 (NOVEMBER 2014)

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5.2.1 ■ ■

■

= 5,45B ■ 1 add

1M

1A (3)

5.2.2 Median = $\frac{R2,76 + R2,91}{2}$ ■

2

= (■

= R2,84 ■ 1M

1S

1A

(3)

5.2.3 Modal value = R6,50 ■■ 2A (2)

5.2.4

■

■ 1M

1A (2)

$$5.2.5 \text{ Range} = R_{3,36} - R_{2,16} \blacksquare \\ = R_{1,20} \blacksquare 2A$$

(2)

$$5.2.6 \text{ } R_{1,39} \blacksquare\blacksquare 2RB (2) \\ [24]$$

TOTAL: 100

NATIONAL
SENIOR CERTIFICATE

GRADE 11

NOVEMBER 2014

MATHEMATICAL LITERACY P2
MEMORANDUM

MARKS: 100

Symbol Explanation

M Method

MA Method with accuracy

CA Consistent accuracy

A Accuracy/Answer

C Conversion

S Simplification

RT/RG /RM Reading from a table/Reading from a graph /Read from map

F Choosing the correct formula

SF Substitution in a formula

J / O Justification

P Penalty, e.g. for no units, incorrect rounding off etc.

R Rounding Off/Reason

This memorandum consists of 7 pages.

2 MATHEMATICAL LITERACY P2 (NOVEMBER 2014)

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QUESTION 1

QUESTION SOLUTION

$$1.1 \text{ } 1.1.1 \text{ } 2 \text{ litres} = 2\,000 \text{ m}^3 \blacksquare \blacksquare$$

$$\text{No of cups} = 2\,000 \text{ m}^3 \blacksquare \\ 125 \text{ m}^3 \blacksquare \blacksquare \\ = 16 \text{ cups} \blacksquare$$

OR

$$125 \text{ m}^3 = 0,125 \blacksquare \blacksquare$$

$$\text{No of cups} = 2 \blacksquare \\ 0,125 \blacksquare \blacksquare$$

= 16 cups ■ 1C Conversion to m■

OR I

1M divided by 125 OR

0,125

1CA

(3)

1.1.2 2 litres = 2 000 m■

No of cups = 2 000 m■

250 m■ ■

= 8 cups ■

OR

250 m■ = 0,250 ■ ■

No of cups = 2 ■

0,250 ■ ■

= 8 cups ■ 1C Conversion to m■

OR I

1M divided by 250 OR

0,250

1CA

(3)

1.1.3 Grade 12A

Profit = Income – Expenses

= (16 x 2,00) ■ – 8,99

= 32 – 8,99 ■

= R23,01 ■

Grade 12 B

Profit = Income – Expenses

= (8 x 3,50) ■ – 8,99 ■

= 28 – 8,99

= R19,01 ■

Disagree , ■ Grade 12A makes R 4,00 more profit (R23,01 – R19,01). ■ 1CA multiplying 16 x 2 according to answer in

1.1.1

1CA difference

between income and

expenses

1CA

1CA multiplying 8 x

3,50

According to answer in

1.1.2

1CA difference

between income and

expenses

1CA

2CA Opinion (8)

$$\begin{aligned} 1.1.4 \text{ Percentage profit} &= \frac{23,01}{8,99} \times 100 \\ &= 255,95 \% \\ &\approx 260 \% \end{aligned}$$

1M concept of %

1A

1CA rounding to
nearest 10 (depending
on answer) (3)

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1.2 1.2.1 Day 3 ■■ 2RG (2)

$$\begin{aligned} 1.2.2 \text{ Difference} &= 88 - 20 \\ &= 68 \text{ cups of cold drink} \end{aligned}$$

2RG correct values

1A (3)

1.2.3 Day 4 ■■ 2RG (2)

1.2.4 60 cups of cold drinks ■■ 2RG (2)

1.2.5 Grade 12 A

Average =

$$\begin{aligned} &\frac{32+60+60+40+72+100+64+80+56+96}{10} \\ &= 660 \\ &\quad 10 \\ &= 66 \text{ cups} \end{aligned}$$

Grade 12 B

Average =

$$\begin{aligned} &\frac{20+40+60+56+44+88+52+60+44+76}{10} \\ &= 540 \\ &\quad 10 \\ &= 54 \text{ cups} \end{aligned}$$

Agree with statement. ■ Grade 12 A's
average is greater than Grade 12B's
average. ■ 1M adding correct
values
1M divided by 10

1CA

1M adding correct
values
1M divided by 10

1CA

2J

(8)

$$\begin{aligned} 1.2.6 \quad 32 : 20 \\ 8 : 5 \end{aligned}$$

1M writing as ratio

1A simplest form (2)
[36]

4 MATHEMATICAL LITERACY P2 (NOVEMBER 2014)
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QUESTION 2

2.1 2.1.1 Number of tins across length

= 48 cm ■

8,4 cm ■

= 5,714... ■

≈ 5 tins ■ 1M using the correct
length

1A find diameter

1A

1R rounding down (4)

2.1.2 Number of tins across width

= 35 cm ■

8,4 cm

= 4,166... ■

≈ 4 tins ■ 1M using the correct
width

1A

1R rounding down (3)

2.1.3 Number of tins stacked on top of each other

= 15 cm ■

3,8 cm

= 3,947... ■

≈ 3 tins ■ 1M using the correct
values

1A

1R rounding down (3)

2.1.4 Number of tins that can go into the box

= 5 x 4 x 3 ■

= 60 tins ■

Vuyo is correct, he will be able to pack more
than 50 tins into the box. ■ 1M

1CA using values as in

2.1.1 – 2.1.3

1CA justification (3)

2.2 Cost of the trip = 7,6 x R 13,20 ■

= R100,32 ■

Post Office Services

1 tin tuna = 170 g

60 tins tuna = 60 x 170 g

= 10 200 g ■

1 000

= 10,2 kg ■

Rate = 35,10 + (4,70 x 10 kg) ■

= 35,10 + 47,00

= R82,10 ■

He must make use of the Post Office Service s
because he will pay R18,22 (R100,32 – R82,10)

less. ■■ 1M
1A

1CA finding the mass
of 60 tins (depending
on answer in 2.1.4)
1A convert g to kg

1CA substitution in
formula

1CA

2CA choose most
economical option (8)
[21]

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QUESTION 3

3.1 3.1.1 South ■■ 2A (2)

3.1.2 One-way streets ■■ 2A (2)

3.1.3 Girton/Park/Jorrisen/Juta/Bre ë/Smit /
Prospect/Olivia/ Park (Any 1) ■■ 2A (2)

3.1.4 B1 OR 1B ■■ 2A (2)

3.1.5 Up Harrison Street ■ right into Bre ë Street
■ left into King George
(Accept any logical directions .) ■■■ 3A (3)

3.1.6 Scale
2,5 cm ■= 500 m
2,5 cm = 500 x 100
2,5 cm = 50 000 cm ■
1 cm = 20 000 cm
1 : 20 000 ■
OR
2,5 cm = 500 m ■
1 cm = 200 m
1 cm = 200 x 100 cm
1 cm = 20 000 cm ■
1 : 20 000 ■ 1A measuring scale on
map

1A converting 500 m to
cm

1A scale

(3)

6 MATHEMATICAL LITERACY P2 (NOVEMBER 2014)
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3.1.7 Marcia

Scale = 1 : 20 000

Distance = 4,2 cm ■ (ACCEPT 4,1 – 4,3)

Distance in meters = $4,2 \times \frac{20\,000}{100}$
= 840 m ■

OR

Distance in meters = 4,2 ■ x 200 (ACCEPT 4,1 – 4,3)
= 840 m ■

Zoe

Scale = 1 : 20 000

Distance = 3,3 cm ■

Distance in meters = $3,3 \times \frac{20\,000}{100}$ (ACCEPT 3,2 – 3,4)
= 660 m ■

OR

Distance in meters = 3,3 ■ x 200 (ACCEPT 3,2 – 3,4)
= 660 m ■

Zoe's ■ distance is 180 m (840 m – 660 m)
shorter than that of Marcia . ■

1A Measuring distance

1CA convert to meters

1A Measuring distance

1CA convert to meters

1CA Identify shortest
distance

1CA difference in
distance

(CA mark applies for
scale calculated in
3.1.6) (6)

3.2 3.2.1 Number of cards in each suite = 52

$$\frac{52}{4} \text{ ■} \\ = 13 \text{ ■ 1M}$$

1A (2)

3.2.2 Probability (black and 10) = $\frac{2}{26} \text{ ■} + \frac{2}{26} \text{ ■}$

$$= \frac{4}{26}$$

$$= \frac{2}{13}$$

$$= \frac{2}{13}$$

13 OR 0,154

OR 15,4% ■ 1A identifying both

1M adding

1A (3)
[25]

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QUESTION 4

4.1 4.1.1 Price before discount = R3 999,90 + R900 ■
= R4 899,90 ■ 1M adding correct
values
1A (2)

4.1.2 % discount = $\frac{900}{4\,899,90} \times 100$ ■
= 18,36772179 ■
= 18,4 % ■ 1M concept of %

1A
1R rounding to
1dec. place (3)

4.1.3 The interest rate is not equal to 8,5 %, but
starts at 8,5%. ■
The repayment value is not equal to
R6 202,39, but starts at R6 202,39. ■
(Accept any other relevant explanation .) 1A: explanation
'from 8,5%'
1A: explanation
'from R 6 202,39'
(2)

4.1.4 $A = P(1 + ni)$
= 3 999,90 (1 + 0,085 x 2) ■
= 3 990,90 (1,17) ■
= R4 679,88 ■

Final answer is not the same. The interest rate
for this purchase is not 8,5%. ■■ 1 SF substitute
correct values
1 S simplify bracket
1CA

2A opinion (5)

4.1.5 ■ You pay much more due to interest
calculated. ■■
OR
■ Item only belongs to you when final
payment is made. ■■
(Accept any relevant answer.) 2A (2)

4.2 4.2.1 Accept any suitable heading . ■■ 2A (2)

4.2.2 Music , ■ videos ■ and photos
(Accept any other usage not mention ed in the
pie chart .) 2A 1 mark per
usage (2)
[18]

TOTAL: 100

